PRODUCTS CATALOGUE



- TEMPERATURE
- HEATER
- METAL CHUCK
- FEEDTHROUGH
- PRESSURE

SMG®
SUNGSIL MFG
주식회사 에스엠지

A Higher Degree of Confidence for SMG Products

㈜에스엠지는 1990년에 설립한 이래 온도센서, 히터, 피드스루 관련 제품을 30여년 이상 생산해 오고 있습니다. 그리고 고객 여러분들께 경쟁력있는 가격의 제품을 공급할 수 있는 제품 생산력과 숙련된 기술경험을 축적해 오고 있습니다.

저희 에스엠지는 고객 여러분들께 보다 나은 기술적 솔루션을 제공하기 위한 끊임없는 노력과 헌신 그리고 기술개발을 사업의 주요 목표로 삼고 있습니다.

모든 제품은 엄격한 품질관리 시스템 절차에 따라 설계, 생산, 관리되며 모든 품질 테스트 요건에 부합될 수 있도록 인증실사 제도를 통해 품질경영 시스템 확립에 필요한 프로그램 개발 및 유지에 힘쓰고 있습니다.

온도센서, 히터, 피드스루 등 다양화된 제품 생산경험과 우수한 인적자원 그리고 최고의 임직원들로 구성된 저희 에스엠지는 고객만족을 위한 광범위한 기술 및 연구개발과 내부혁신에 최선을 다할 것 입니다.

앞으로도 에스엠지에 변함없는 애정과 성원을 부탁합니다.

감사합니다.



COMPANY HISTORY

1990's

1990 성실전자 설립
1993 대형 MOTOR용 RTD 출시
1995 발전소 발전기용 RTD 출시
1999 성실전자 주식회사 법인 전환

2000's

2001 RTD UL 인증 취득 (FX type)
2002 수출 유망 중소기업 선정
2004 클린사업장 선정
2004 ILLICA Group 업무 제휴 (미국)
2004 BLUE PETER Series® 브랜드로 미주 공략
2005 성실전자(주) 기술부설연구소 설립
2006 RTD BV 선급인증 획득
2007 화성 공장 준공 및 이전
2008 Epoxy, Kapton CE 인증 획득, Mica Heater 개발
2009 가산디지털단지 연구소 및 영업소 설치
RTD KTL 국내방폭 인증 획득 (PT100, PT1000)

2010's

2010 Cable Bushing 개발 및 IECEx, ATEX 방폭 인증 획득
2011 RTD ATEX 방폭 인증 획득 (PT100, PT 1000)
2012 TX Type RTD 개발 판매, ISO 9001, ISO 14001 인증 획득
2013 SUNGSIL, SMG 상표등록 출원, 실리콘 리본 히터 개발
2014 접착식 Mica Heater, Kapton, 실리콘 히터 특허출원 진행
드로잉 타입 Kapton Heater 중국 실용신안 취득
㈜SMG 법인명 변경
2015 ㈜에스엠지 신사옥 준공 및 이전 완료
대형 발전기용 Glass Bushing 개발 완료
2016 GE (General Electric) Partner Company
이노비즈 기술혁신형 중소기업 선정
한국방위산업진흥회 회원등록
2018 경기도 유망중소기업 선정
2019 GB Series 피드스루 국산화

2020's

2020 수출유망중소기업 지정 - 중소벤처기업부
RTD 국내방폭 인증 획득 (PT100, PT1000) KGS Ex ia (BTR51),
WIKA 판매대리점 계약, Honeywell 판매대리점 계약
2021 5대발전소 기자재 유자격 등록
RTD (BTR51) BV 선급인증 획득
2023 시즈타입 센서 방폭인증 획득 (Ex db, Ex ia, Ex e)
2024 시즈타입 센서 방폭인증 획득 (Ex tb)

CONTENTS

04 TEMPERATURE SENSOR
FX, TX Stator winding RTD
BTR51 Coil End Temperature Detector
STD Surface Temperature Detector
STR Industrial Process Resistance Temp.Detector(RTD)
STC Industrial Process Thermocouples(TC)
STS Industrial Process Thermostats
STW Industrial Process Thermowells

26 METAL CHUCK
METAL CHUCK HEATER
METAL HEATER
AIR HEATER

32 HEATER
HK Kapton Film Heater
HC Clear view Heater
HM, HMC Mica Plate Heater
HE Epoxy Plate Heater
HS Silicone Rubber Heater

38 방산부품(히터) LIST

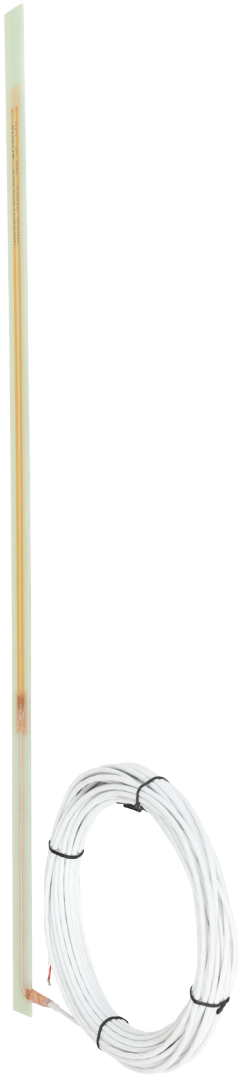
40 CABLE BUSHING
CB, CBC Power, Sensor, and other lines
GB Power, Sensor(RTD/TC)
MB Power

46 PRESSURE SENSOR
SPT Industrial Pressure Transmitter

48 Technical Reference, PT 100Ω 저항값과 온도 비교표, PT 1000Ω/NI 120Ω 저항값과 온도 비교표, CU 10Ω 저항값과 온도 비교표

FX, TX

Stator Winding RTD



FEATURES

모터의 과열 방지를 감지하기 위해 고정자의 슬롯 사이에 삽입하여 사용한다.
백금, 구리, 니켈 등을 사용하는 센서로서 두께가 얇고 무유도성이며 진동에 강하고 고압에서 견딜 수 있는 구조로 설계되었다.
센서 길이 전체가 감온부이므로 평균값을 측정하여 한 부분만 측정해서 생기는 오류를 막을 수 있다. 또한 다양한 형태의 주문형 제품의 생산이 가능하다.

Embedded temperature monitoring of motors and generators is a long time industry accepted practice, allowing for continuous assessment of equipment condition. The embedded method of monitoring utilizes specially designed Resistance Temperature Detectors(RTD) and Thermocouples(TC) placed outside the major insulation as specified in ANSI C50-10-1990, Section 5.2.
SMG's temperature sensors are designed specifically to accurately sense and respond to temperatures of the windings in motors and generators. This allows constant monitoring to ensure equipment operation within design limits.
The sensing element of SMG's RTD extends through most of the body length. The sensor design allows for averaging the measurement process over a greater portion of the slot length. This eliminates the problems related to tip sensitive sensors that are inches in body length, this is a useful characteristic.

APPLICATION

중형·대형 모터, 발전기, 그 외 다수 사용처
Large Motors and generators, etc.

CERTIFICATION

Explosion proof certification

- ATEX Ex ia IIC T6...T3 Ga
- ATEX Ex eb IIC Gb
- IECEx Ex eb IIC Gb
- IECEx Ex ia IIC T6...T3 Ga
- KTL Ex e IIC
- KGS Ex ia IIC T6...T3

Ship's classification

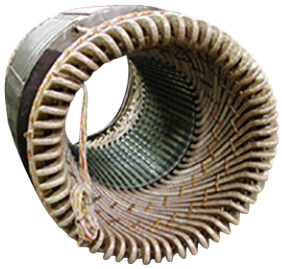
- BV 18688/C0 BV



INSTALLATION

고정자 온도검출 센서는 중대형 모터와 발전기의 슬롯 사이에 설치한다.
모터당 6개의 센서 설치가 권장되며 상 당 2~3개씩 설치한다.
최적의 성능 수행을 위해 가장 열이 많이 올라가는 부분에 설치한다.

Stator winding sensors fit on the coil end of large motors and generators.
Six sensors are recommended for each motor, two per phase.
Locate sensors near the hottest point of the windings for best performance.



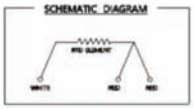
STANDARD

IEC 60751
JIS C 1604

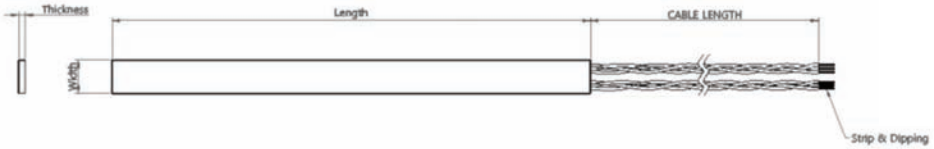
TYPE OF STATOR WINDING SENSOR

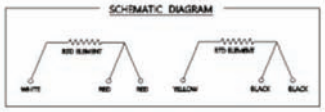
FX(Single) Type



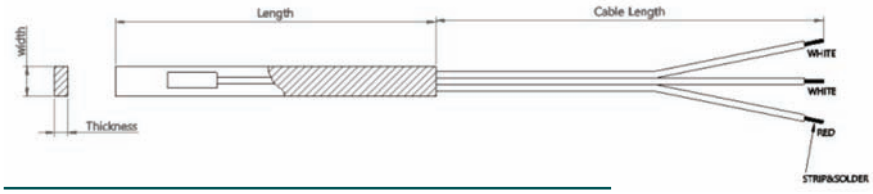


FX(Dual) Type





TX(Single&Dual) Type



No.	Number of Element	Schematic Diagram & Wire Color
1	3(Single)	
2	4(Single)	
3	6(Dual)	
4	8(Dual)	

Note. SMG can be custom-made according to the field process.

ORDERING CODE

1. Model No.

WTD	Stator Winding RTD / TC
	FX_ Drawingtype(ClassF-155℃)
	FXH_ Drawingtype(ClassH-180℃)
	TX_ Chiptype(ClassF-155℃)
	TXH_ Chiptype(ClassH-180℃)

2. Body Length(mm)

0	155 (STANDARD)
1	Etc (Min. 30mm ~ Max. 5,000mm) (TBD)

3. Body Thickness(mm)

0	1.3
1	2.0
2	Etc (TBD)

4. Body Width(mm)

0	6 (Single type)
1	11 (Dual type)
2	12(Dual type)
3	Etc (TBD)

5. Type of Element

0	Pt100ohm at 0℃
1	Pt1000ohm at 0℃
2	Copper 10Ω at 25℃
3	Copper 10Ω at 0℃
4	Ni 120Ω at 0℃
5	TC K type (J, E, T etc.)

6. Numberof Element

0	Single (2Wire)
1	Single (3Wire)
2	Single (4Wire)
3	Dual (6Wire)
4	Dual (8Wire)

7. Leadwire Length (M)

0	4
1	6
2	8
3	Etc (TBD)

8. Leadwire Insulation

0	Teflon wire (STANDARD)
1	Silicon shield wire
2	Teflon shield wire

9. Tolerances on Temperature Reading

0	Class A
1	Class B
2	Etc (TBD)

10. Explosion proof Certificate

0	None
1	ATEX Ex eb
2	ATEX Ex ia
3	IECEx Ex eb
4	IECEx Ex ia
5	KTL Ex e
6	KGS Ex ia
7	BV Marine

SAMPLE CODE

WTD FX-0-0-0-0-1-0-0-0-3

BTR51

Coil End Temperature Detector



FEATURES

The thin film sensor element or cylindrical sensor element is wrapped with a heat-resistant shrinkable tube, and the inside is injected and filled with liquid silicon, so it is designed to have a fast response speed and be resistant to vibration.
WTD type consists of elements assembled into shrinkable tubes.

- Small, rugged RTD's and Thermocouples
- Excellent vibration resistance
- Fast responsibility

APPLICATION

Motors and Generators, etc.

CERTIFICATION

Explosion proof certification

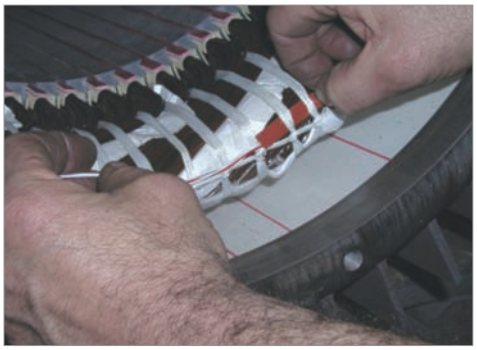
- ATEX Ex ia IIC T6...T3 Ga
- KGS Ex ia IIC T6...T3

Ship's classification

- BV 18688/C0 BV

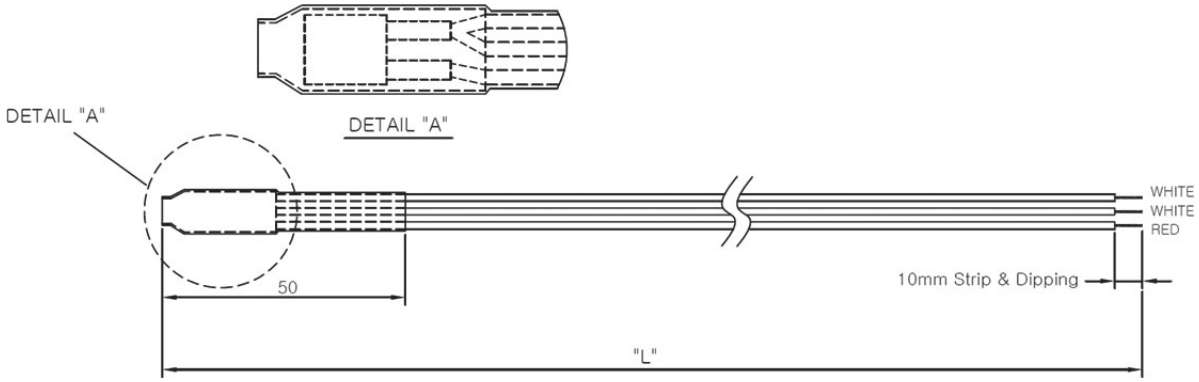
INSTALLATION

Installed in the end of coils to large motors and Generators.



PRODUCT OVERVIEW

BTR51 type detector is inserted in the end of the coil to prevent the over heating of the motors and generators detector element materials are flexible, non-inductive, stable at vibration.
The selecting element of BTR51 type can be designed and manufactured in a variety of different shapes. BTR51 type minimizes the risk of draining a biased result by accurate temperature measured along the wanted spot.



ORDERING CODE

1. Element Type

PT100B	Platinum 100Ω at 0℃ ± 0.3%
PT100A	Platinum 100Ω at 0℃ ± 0.15%
PT1000B	Platinum 1,000Ω at 0℃
K	T/C K type (J, E, T etc.)

2. Lead wire length

3M	3 Meter (Lead wire length in meters)

3. Number of lead wire

0	2 wire(Single)
1	3 wire(Single)
2	4 wire(Single)
3	4 wire(Dual)
4	6 wire(Dual)
5	8 wire(Dual)

4. Lead wire insulation

T	Teflon wire
S	Silicon shield wire

5. Explosion proof Cert.

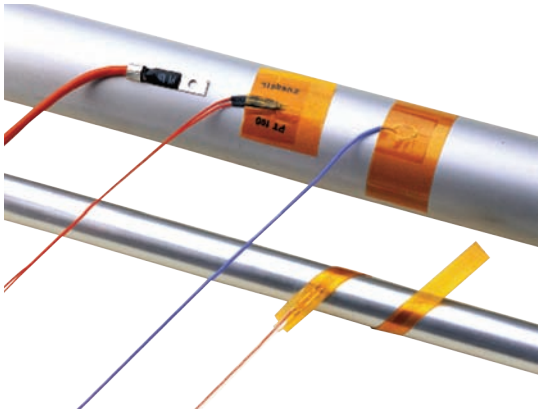
N/A	Non-explosion (Standard)
Ai	ATEX Ex ia
Ki	KGS Ex ia
B	BV Marine

SAMPLE CODE

BTR51	PT100B		3M		3		T		Ai
	1	-	2	-	3	-	4	-	5

STD

Surface Temperature Detector



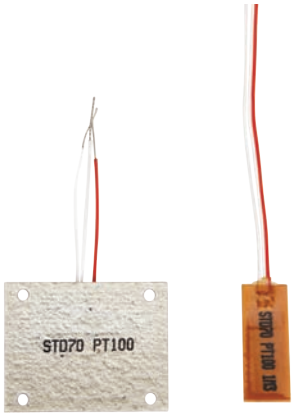
FEATURES

배관파이프, 탱크, 충전지, 고압트랜스 등의 표면온도를 측정하고 내부액체 또는 코일 등의 과열 또는 동결로 인한 소손을 방지하기 위해 사용된다. 내부에 백금센서(또는 써머커플)를 탑재한 온도센서로써 감지속도가 빠르고 소형이며 습기에 강하다.

- 다양한 형태의 표면에 빠르고 쉽게 부착할 수 있다.
- Adhere to all types of surfaces for quick and easy mounting.

길이 전체의 평균값을 측정함으로서 부분측정의 에러를 제거할 수 있다. 곡면에 취부할 수 있다. 모터와 발전기의 권선온도를 측정하도록 특별히 디자인 되었으며 권선 끝부분에 취부 할 수 있다.

Average temperatures along their length to eliminate point measurement errors. Wrap them around cylinders or adhere them to flat surfaces. Specially designed to sense stator temperatures in motors and generators. Mounts on the end of stator windings instead of inside them Use it for motors, generators or retrofit.



APPLICATION

모터의 코일앤드, 파이프배관의 표면, 충전용 배터리의 표면, 액체 고체탱크의 표면, 항공우주 산업, 군사 장비와 의료 장비 등

End of stator windings, Surface of pipe, battery, liquid tank, Aerospace, Military equipment and medical equipment etc.

INSTALLATION

STD's lend themselves to a variety of installation methods. You should avoid repeated bending during the installation process, and not flex in use unless they are specifically designed to do so. Take care to secure leadwires so they do not pull against sensor bodies. Leadwires should be routed along the sensed surface a short distance so that they do not sink heat away from the sensing element.

USED AND ONSTRUCTION

표면 온도센서는 다양한 설치방법을 제공한다. 설치 중에 반복해서 굽히지 말아야 한다. 휘어지도록 설계되었지만 반복적 휨이 계속되면 제품에 무리가 갈 수 있다. 몸체 밖으로 빠지지 않도록 리드 와이어를 잡아당김에 주의하여야 한다. 선은 센서 표면에서 짧은 거리만 따라나오고 우회하여 서 열로부터 분리되어야 한다.

Platinum is the most widely used material for RTDs. This is due to its stable and predictable nature. it has the highest useful temperature range(up to about 300°(with special construction), is the most repeatable at higher temperature and withstands oxidation / corrosion Cost considerations may dictate the use of copper and nickel, but these offer a relatively low range of temperature.

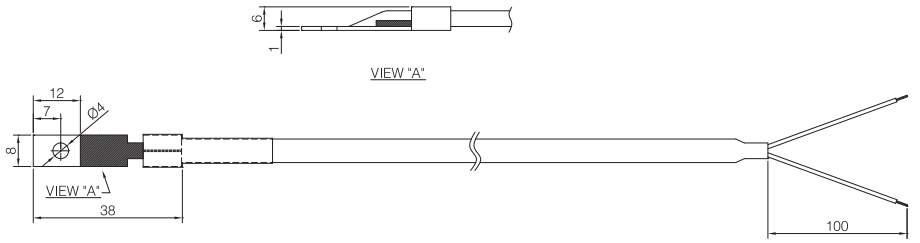
RTD CHARACTER-ISTICS

Resistance Temperature Detectors(RTDs) are devices that sense temperature by measuring the change in resistance of a material in a known and repeatable manner. An RTD usually contains either a conductive film or a coil of wire. Extension wires allow the electrical resis tance to be measured from some distance. RTD elements are packaged in various configurations according to the desired application. RTDs are preferred when accuracy, long-term stability and repeatability are important in an application. Note these specifications are typical, consult the factory for specific performance based on model number.

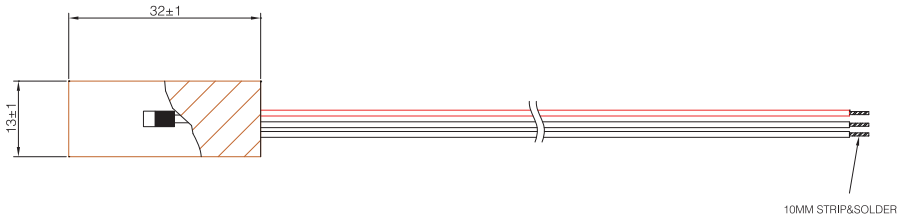
SPECIFICATION

	Sensor Element	Operating temperature	Connection method	Process connection
STD30	PT100, 1×PT1000	-62℃ ~ 200℃	Single(2, 3, 4) Dual(4, 6, 8)	Screw mounting(4Ø)
STD50	PT100(TFS), PT1000	-55℃ ~ 180℃		-
STD60	PT100(Drawing type)	-55℃ ~ 180℃		-
STD70	PT100, PT1000	-55℃ ~ +260℃		-

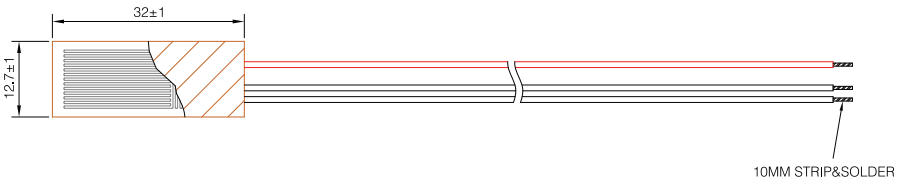
STD30



STD50, 70



STD60



ORDERING CODE

1. Model Type

STD30	Screw mounting type (hole Ø4)
STD50	Kapton film type + TFS (option : PSA)
STD60	Kapton film type + Pt drawing (option : PSA)
STD70	Kapton film type + TFS

2. Element

Pt100	Platinum 100℃ at 0℃
Pt1000	Platinum 1000℃ at 0℃
TC	Thermocouple (K. E. T...)

3. Lead wire length

3M	3 Meter (Lead wire length in meters)

4. Number of lead wire

S2/S3/S4	2, 3, 4 (Single type)

SAMPLE CODE

STD60	PT100	3M	S3
1	2	3	4

STR

Industrial Process Resistance Temp.Detector(RTD)

INTRODUCTION

Electrical resistance of any metallic conductor varies according to temperature changes. The sensor for measurement of temperature by utilizing this phenomenon is called “Resistance Temp.Detector” or “RTD” and can measure temperatures more accurately than other temperature sensors.
Except the end tip length of 50 mm, it could be bent 5 times more than its out diameter so can be used in the process pipe where it is not easy to install due to the obstacles that exist around the installation process.

Certificate (Explosion Proof Head Type)

- HEAD : IECEx/ATEX(CE)/KGS Ex db eb IIC T6
- NONE HEAD : IECEx/ATEX(CE)/KGS Ex ia IIC T6···T4
  IECEx/ATEX(CE)/KGS Ex eb IIC T6



WEATHER PROOF



EXPLOSION PROOF



COMPACT/DIN TYPE



NONE HEAD TYPE

STANDARD FEATURES

Diameter

3.2, 4.8, 6.0, 6.4 and 8.0 mm

Element

Pt100ohm at 0℃, Pt1000ohm at 0℃, JPt100ohm at 0℃

Tolerances on Temperature Reading

Class A, Class B(DIN/IEC751-1, BS/EN60751-1, JIS C1604)

Features

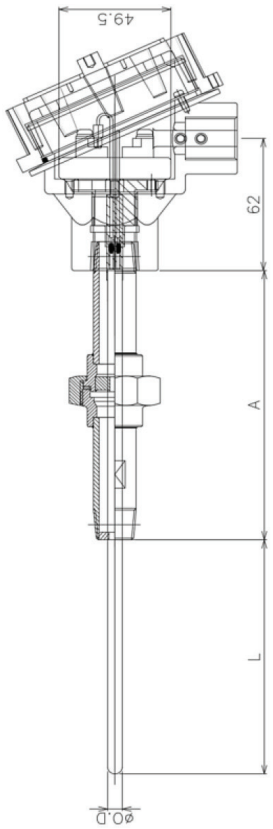
- Quick Response, High Flexibility, High Accuracy
- Wide Variety of Specifications
- Wide Range of Measuring Temperatures (-200 ~ 650℃)

Head material

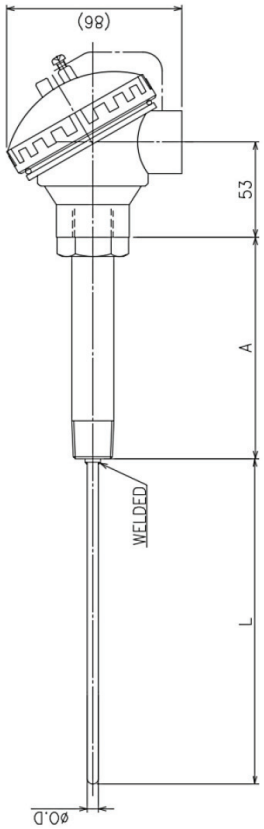
- ALDC (Standard)
- 304SS (Not available compact type)

TYPE OF RESISTANCE TEMP.DETECTOR

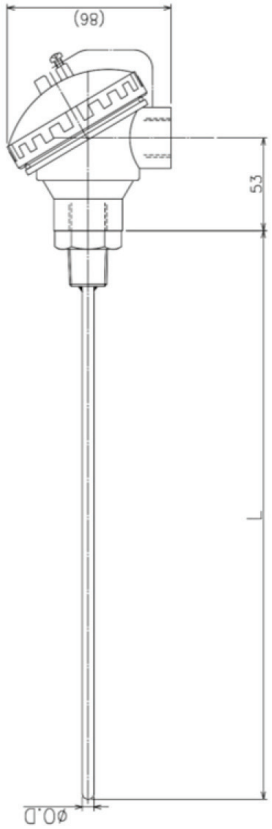
Explosion Proof, Union & Nipple



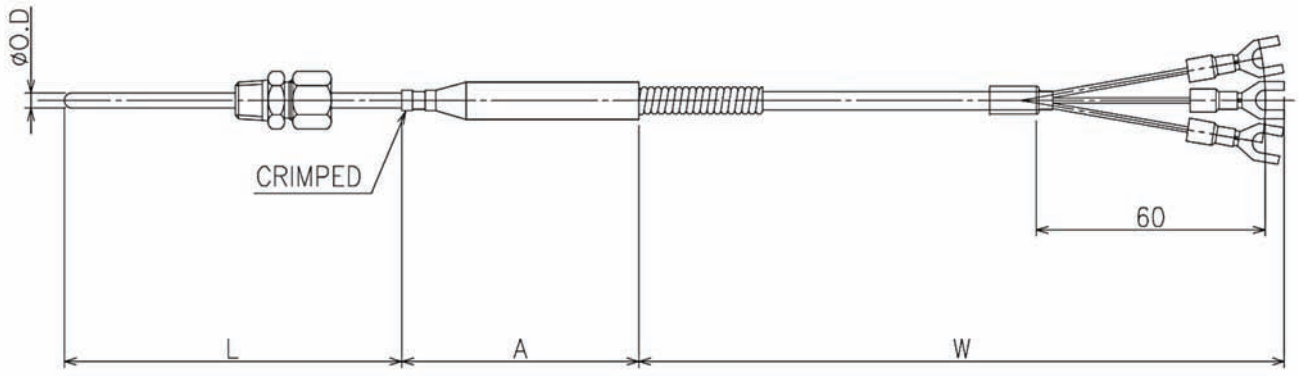
Weather proof, Nipple



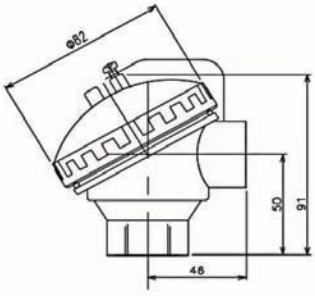
Weather proof, Thread



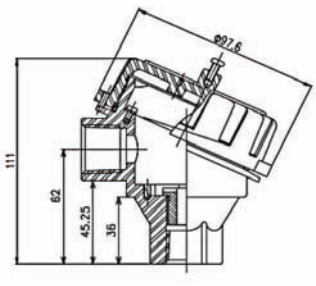
None Head Type w/CAm.Fitting



General Type(Weather Proof, IP68)



Explosion Proof Type(IP68)



Note. SMG can be custom-made according to the field process.

ORDERING CODE

1. Model No.

STR	Industrial Process Resistance Temp.Detector
	EX_Explosion Proof (Head, Ex d, IP 68)
	EX N_Explosion Proof (None Head, Ex ia)
	GW_Weather Proof (IP68)
	DC_Din Connector Type
	N_None Head Type

2. Head material

0	ALDC
1	SUS (Not available compact type)

3. Cable Entry(Head Type)

0	PF 1/2" (STD)
1	PF 3/4"
2	NPT 1/2"
3	NPT 3/4"
4	Other

4. Sensor Diameter

0	3.2
1	4.8
2	6.0
3	6.4
4	8.0

5. Number of Element

0	Single (3Wire)
1	Single (4Wire)
2	Dual (6Wire)
3	Dual (8Wire, Over 6.4mm)

6. Type of Element

0	Pt100ohm at 0℃
1	Pt500ohm at 0℃
2	Pt1000ohm at 0℃

7. Sensor Material

0	316SS
1	316LSS

8. Type of Mounting

0	None
1	Nipple (1/2B, 100L, 304SS)
2	Union & Nipple (1/2B,150L, 304SS)
3	Thread (304SS, 316SS)
4	Other

9. Tolerances on Temperature Reading

0	Class A (JIS A)
1	Class B (JIS B)
2	Class AA (JIS AA)

10. Process Connection

0	None
1	1/4"
2	1/2"
3	3/4"
4	1"

11. Process Connection Type

0	None
1	PF(G)
2	PT
3	NPT
4	CF+PT (Com.Fitting)
5	CF+NPT (Com.Fitting)
6	Other

12. Insert Length (mm)

XXX	Other

13. Insert Type

0	None
1	Spring Loaded

14. Lead Wire

0	TBD

15. Accessories

0	None
1	1 With Thermowell

SAMPLE CODE

STR GW-0-0-3-0-0-1-2-0-3-3-200-1-0

SMG®

STC

Industrial Process Thermocouples(TC)

INTRODUCTION

It is a temperature sensor that is widely used for industrial thermometry from low temperature to ultra high temperature using precious and non-metallic wires. We manufacture products for various purposes, such as general industrial thermocouples and explosion-proof thermocouples used in dangerous places of explosion.It is a sheath thermocouple with insulation and high pressure resistance that is tightly filled with high-purity magnesia powder between metal sheath and thermocouple wires.

Certificate (Explosion Proof Head Type)

- HEAD : IECEx/ATEX(CE)/KGS Ex db eb IIC T6
- NONE HEAD : IECEx/ATEX(CE)/KGS Ex ia IIC T6···T4
  IECEx/ATEX(CE)/KGS Ex eb IIC T6



WEATHER PROOF



EXPLOSION PROOF



COMPACT TYPE



NONE HEAD TYPE

STANDARD FEATURES

Diameter

1.0, 1.6, 2.3, 3.2, 4.8, 6.0, 6.4 and 8.0 mm

Element

ISA R, S, B, K, J, E, T, N and C(W5)

Tolerances on Temperature Reading

Class 1, Class 2(DIN/IEC584-2, BS/EN60584-2, JIS C1602)

Hot junction

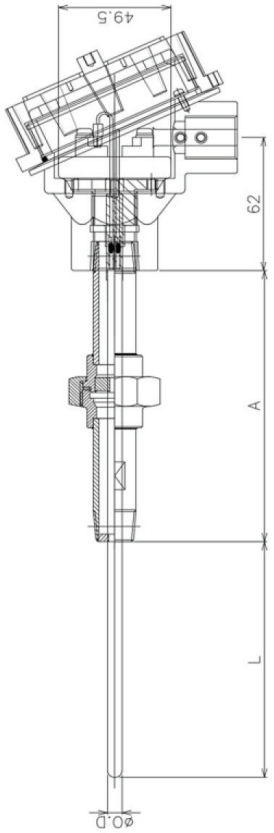
- Ungrounded / Grounded / Exposed

Head material

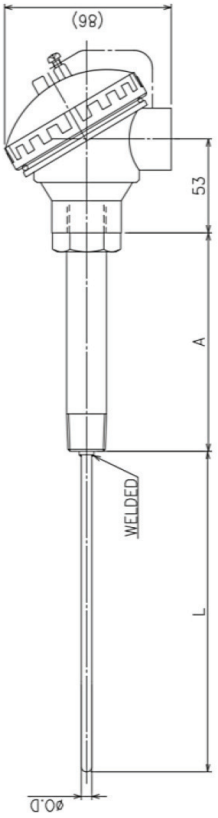
- ALDC (Standard)
- 304SS (Not available compact type)

TYPE OF THREMOCOUPLE

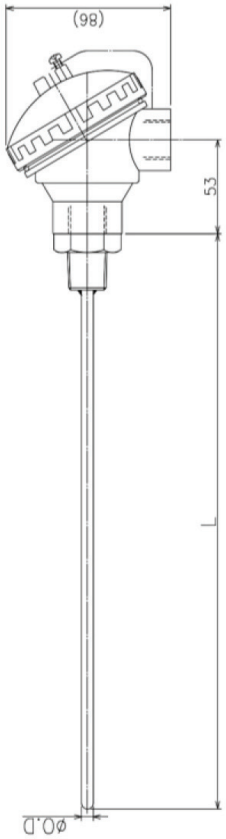
Explosion Proof, Union & Nipple



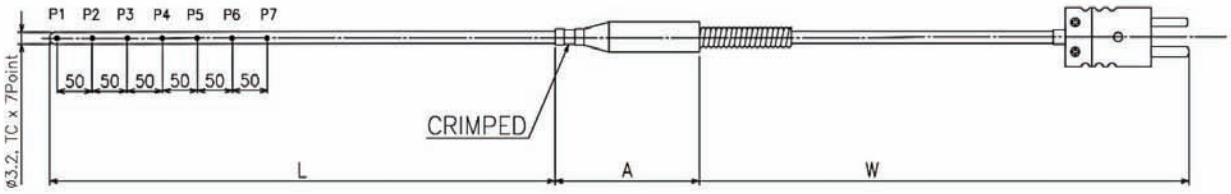
Weather proof, Nipple



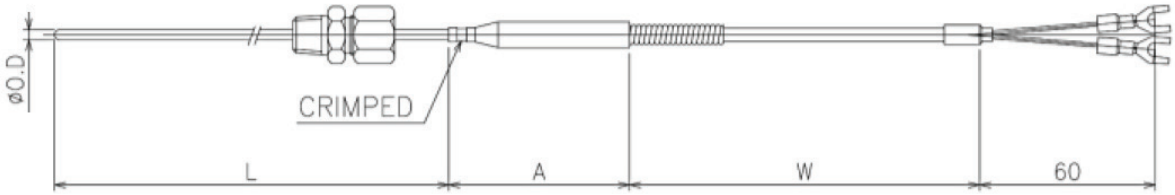
Weather proof, Thread



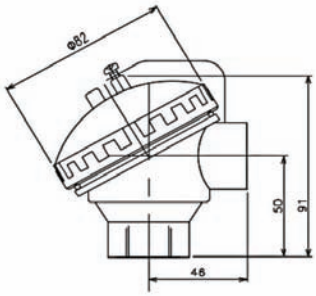
None Head Type w/Quick Connector



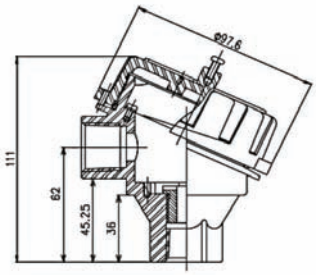
None Head Type w/Com.Fitting



General Type(Weather Proof, IP68)



Explosion Proof Type(IP68)



Note. SMG can be custom-made according to the field process.

ORDERING CODE

1. Model No.

STC	Industrial Process Thermocouples
	EX_Explosion Proof (Head, Ex d, IP 68)
	EX N_Explosion Proof (None Head, Ex ia)
	GW_Weather Proof (IP68)
	N_None Head Type

2. Head material

0	ALDC
1	SUS (Not available compact type)

3. Cable Entry(Head Type)

0	PF 1/2" (STD)
1	PF 3/4"
2	NPT 1/2"
3	NPT 3/4"
4	Other

4. Sensor Diameter

0	1.0 (Available Only Single)		
1	1.6	5	6.0
2	2.3	6	6.4
3	3.2	7	8.0
4	4.8	8	Other

5. Number of Element

0	Single (2Wire)
1	Dual (4Wire)
2	Tripple (6Wire) From 3.2mm)
3	Other (TBD)

6. Type of Element

0	R	4	E
1	S	5	J
2	B	6	T
3	K	7	N

7. Sensor Material

0	316SS
1	316LSS
2	310SS
3	INCONEL 600
4	Other (TBD)

8. Tolerances on Temperature Reading

0	Class 1 (JIS 1)
1	Class 2 (JIS 2)
2	ASTM

9. Hot Junction

0	Ungrounded
1	Grounded
2	Expososed

10. Process Connection

0	None
1	1/4"
2	1/2"
3	3/4"
4	1"

11. Process Connection Type

0	None
1	PF(G)
2	PT
3	NPT
4	CF+PT (Com.Fitting)
5	CF+NPT (Com.Fitting)
6	Other

12. Type of Mounting

0	None
1	Nipple (1/2B, 100L, 304SS)
2	Union & Nipple (1/2B, 150L, 304SS)
3	Thread (304SS)
4	Other

13. Insert Length (mm)

XXX	Other

14. Insert Type

0	None
1	Spring Loaded

15. Lead Wire

0	TBD

16. Accessories

0	None
1	With Thermowell

SAMPLE CODE

STC GW-0-0-6-1-3-0-0-0-2-3-2-200-0-0

SMG®

STS

Industrial Process Thermostats



INTRODUCTION

Temperature switches are generally used in industry for limiting temperature. They monitor the temperature of machinery and equipment and, for example, switch off machinery if it overheats or switch on a fan to cool the equipment. Bimetals form the basis of the SMG temperature switches model STS Series.

Features

- Snap Action, Automatic Reset
- Superior Heat Response
- PBT Resin Case with Epoxy Sealing
- Wire Customization upon Request
- IP(Ingress Protection): IP45 (Solid 4, Liquid 5)
- Normally Closed

Applications

- Electric Motors, Battery Chargers, Transformers
- Power Supplies, Heating Pads, Fluorescent Lamp, Ballasts
- OA-Machines, Solenoids, LED Lighting, etc.

STANDARD FEATURES

Heat Endurance

- Open Temp. +50℃/continuity, 200℃/1min

Temperature Setting Range

- 50℃ ~ 150℃ in increments of 5℃

Dielectric strength

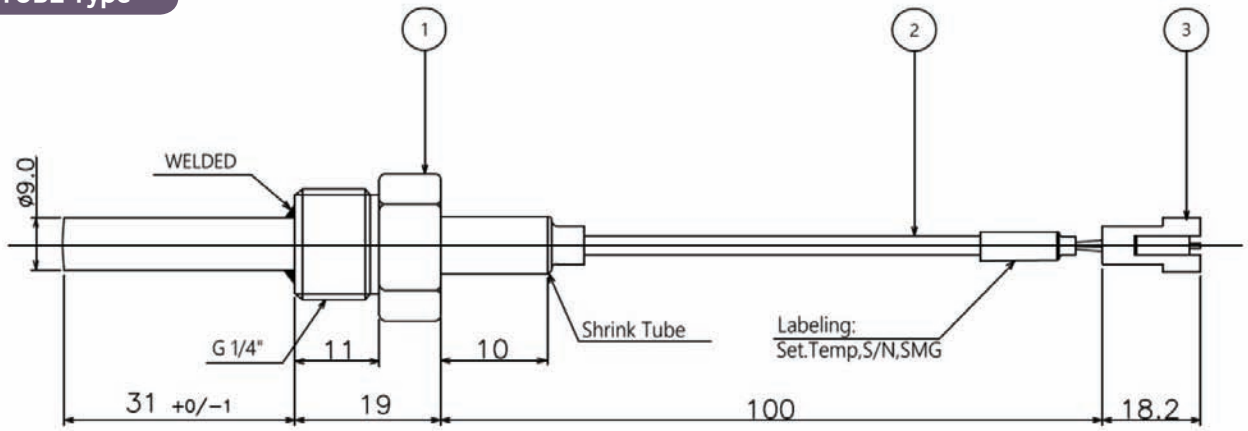
- AC 1500 V for 1min

Service Life

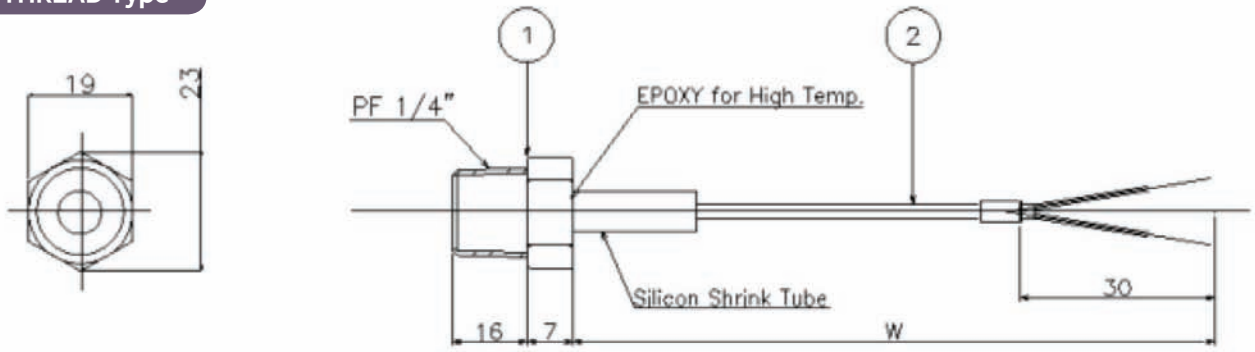
- DC 48V/3A, 6,000 Cycle (According to UL 873)

DIMENSION

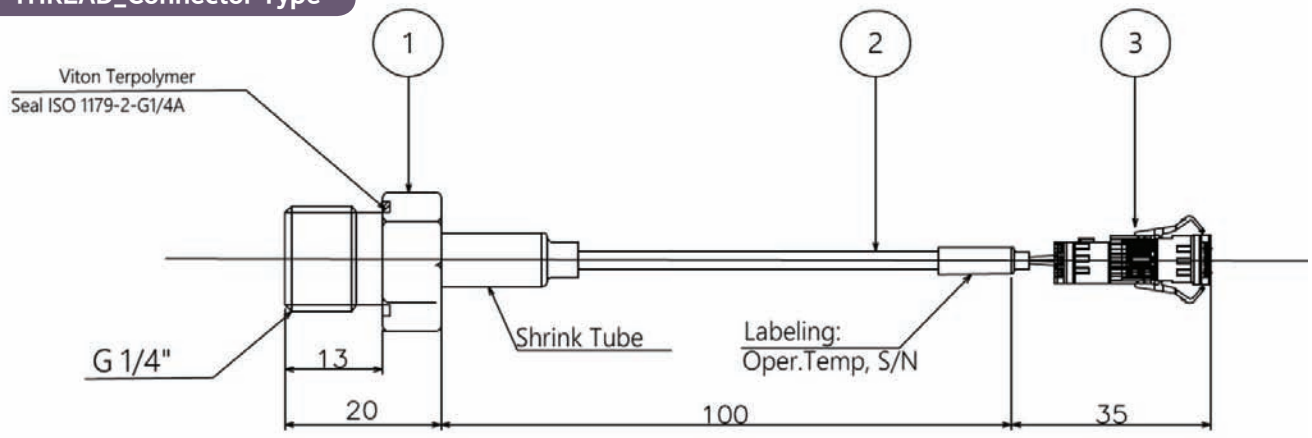
TUBE Type



THREAD Type



THREAD_Connector Type



Note. SMG can be custom-made according to the field process.

ORDERING CODE

1. Model No.

STC	SMG Industrial Process Thermostats
	N_Tube Type
	T_Thread Type

2. Sensor material

0	304SS
1	316SS
2	Cu

3. Sensor Diameter (mm)

0	9.0
1	10.0
2	12.0
3	15.0
4	Other

4. Insert Length (mm)

XXX	Custom-made

5. Number of Element

0	Single (2Wire)

6. Tolerances on Temperature Reading

0	SV ± 5℃

7. Temperature Setting Range (℃)

0	50
1	80
2	100
3	110
4	120
5	130
6	140
7	150
8	Other

8. Process Connection

0	None
1	1/4"
2	1/2"
3	3/4"
4	1"

9. Process Connection Type

0	None
1	PF(G)
2	PT
3	NPT
4	Other

10. Wire Material

0	SILICON_0.3SQ-2CORE
1	Other

11. Wire Length (mm)

0	100
1	Custom-made

12. End Terminal

0	None
1	Pin Connector
2	Quick Connector
3	Other

SAMPLE CODE

STS-T-0-0-30-0-0-5-0-0-0-0-1

SMG®

STW

Industrial Process Thermowells



INTRODUCTION

Thermowells which are made of solid bar stock of various heat and corrosion resistant alloys by drilling are usually preferred tubes for critical applications where high mechanical strength and longer services life are required.

If the alloy bar material is correctly selected and designed properly, the Thermowell lasts long against corrosives, high pressure, high temperature, mechanical shock and vibration that may result from high velocity of fluids.

In order to offer the best and safest Thermowells against Ka 'rma'n's Turbulence and other stresses, automatic calculations of mechanical strength to fluid pressure and flow velocity to estimate frequency of critical resonance are made by a specially developed compluter program based upon operating conditions at the site.

Thermowells are manufactured by a genuine Two-Shaft Gun Drilling Machine of 2-metre max. depth and the latest NC Turning Machines.

STANDARD FEATURES

Round Bar (O.D, mm)

- 25, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50 and 55

Standard Alloy Bar Materials

- Stainless Steel series : 304SS, 321SS, 316SS/LSS, 347SS, 310SS, 446SS, 253MA
- Inconel series : 600, 601, 625, 825
- Hastelloy series : Hast-C, Hast-X, Hast-C276
- Other : Monel600, Brass, HCF, 50Co-30Cr, Bronze, Titanium, Tantalum, Molybdenum

Test and Inspections

- Pressure Test : N. glass pressure test up to 10MPa is conducted upon request
- Hydrostatic Pressure Test : Internal pressure test up to 40MPa is conducted upon request
- X-Ray Inspection(RT) : X-ray inspection for uniformity in wall thickness, eccentricity of bore and smooth inner finish are also conductive upon request

Optional:

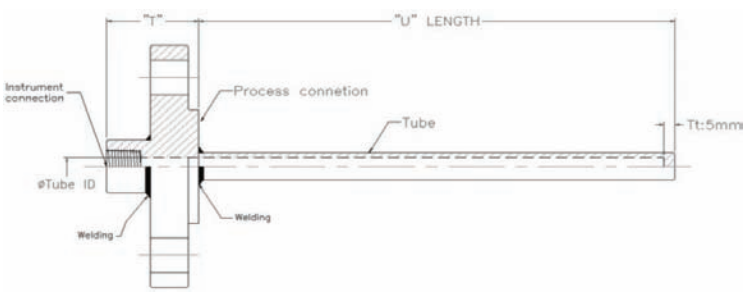
- Helium leak Test(H2)
- Dye penetrant Test(PT Test)
- Cross checking of material with Mill Certificate

Hex Bar (mm)

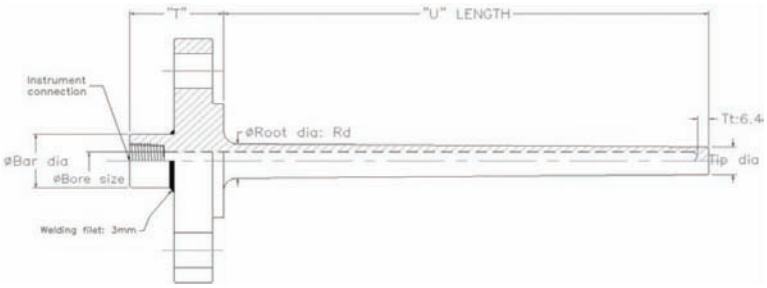
- 27, 32, 35, 38 and 41

TYPE OF THERMOWELLS

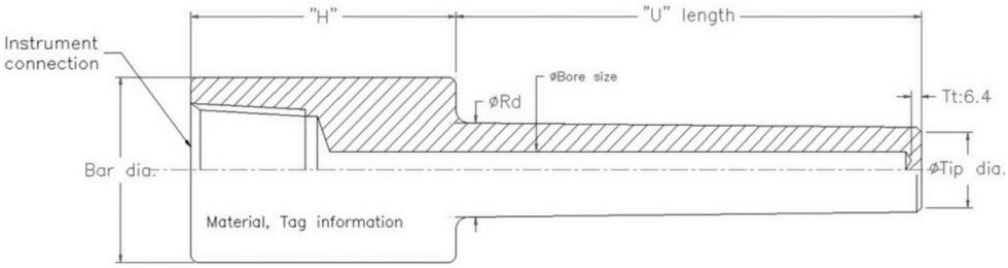
Drilled bar Stock (Taper / Straight) Flanged type_STW DF



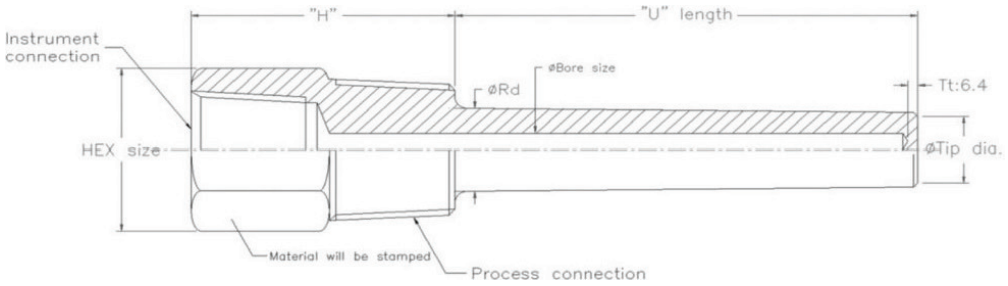
End Closed Pipe Flanged type_STW PF



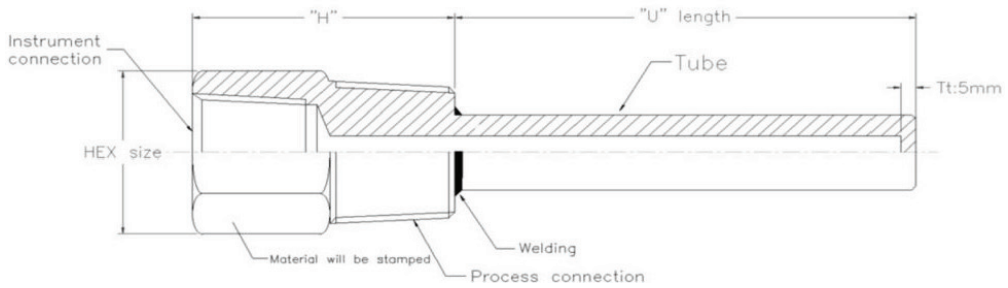
Drilled bar Stock (Taper / Straight) Socket Weld in Type _STW DW



Drilled bar Hex Thread (Taper / Straight) Type_STW DH



End Closed Pipe Hex Thread Type _STW PH



ORDERING CODE

1. Model No.

STW	Industrial Process Thermowell
	DF_DriledbarStock(Taper/Straight) Flanged type
	PF_EndClosedPipeFlangedtype
	DW_DrilledbarStock(Taper/Straight)SocketWeldinType
	DH_DrilledbarHexThread(Taper/Straight)Type
	PH_EndClosedPipeHexThreadType

2. Material of Well

0	304SS
1	304LSS
2	316SS
3	316LSS
4	310SS
5	321SS
6	INCONEL600
7	MONEL400
8	HAST-C
9	A182F304
10	Other

3. Internal Connection

0	1/2"
1	1 3/4"
2	2 1"

4. Internal Connection Type

0	PF(G)
1	PT
2	NPT
3	XXX

5. Insert Length “U-Length” (mm)

XXX	Other(TBD)

6. Insert Length “T-Length” (mm)

0	45
1	50
2	70
3	Other(TBD)

7. Root / Tip / Bore Size (mm)

xxx	xx/xx/x
For Tapered	ex) 21 / 17 / 7
For Straight	ex) 17 / 17 / 7

8. Process Connection (Flanged type)

0	1/2"	6	10A
1	1"	7	15A
2	1-1/2"	8	20A
3	2"	9	25A
4	2-1/2"	10	40A
5	3"	11	Other

9. Process Connection Type (Flanged type)

0R(F)	ANSI 150# RF(FF)
1R(F)	ANSI 300# RF(FF)
2R(F)	ANSI 600# RF(FF)
3RJ	ANSI 600# RTJ
4R(F)	ANSI 900# RF(FF)
5R(F)	JIS 10K RF(FF)
6R(F)	JIS 20K RF(FF)
7R(F)	JIS 30K RF(FF)
8	Other

10. Process Connection

0	1/2"
1	3/4"
2	1"
3	1"SW
4	2"SW
5	Other

11. Process Connection Type

0	PF(G)
1	PT
2	NPT

12. Accessories

1	Plug & Chain(304SS)

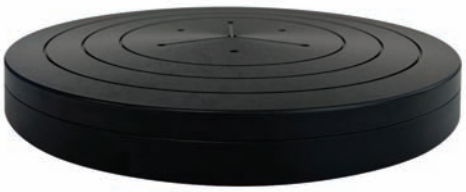
Note. SMG can be custom-made according to the field process.

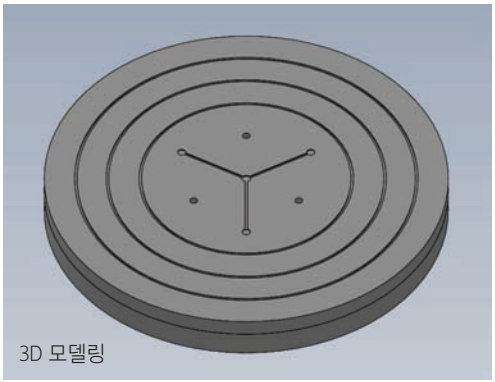
SAMPLE CODE

STW DH-2-0-2-150-0-0-1-2-1

SMG®

METAL CHUCK HEATER





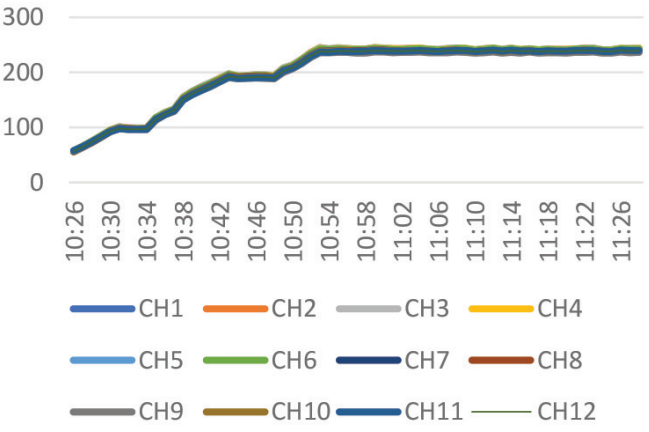
3D 모델링

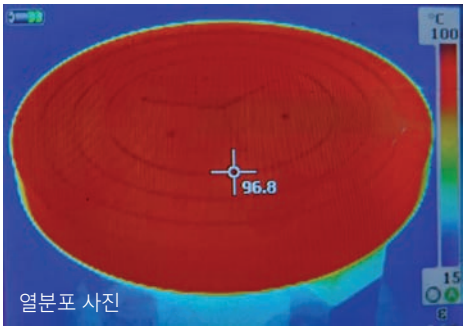
SPECIFICATIONS

Operation Temp	-50℃ ~ 350℃ / Changeable
Production Size	8" - 12" / Changeable
Thickness	35t / Changeable
Input Voltage	AC, DC
Option	Anodizing, Suction, Sensor, Plating etc
Quality of the material	Aluminum, Oxygen Free Copper, MMC
Heater	MICA Heater Excellent temperature stability compared to conventional sheath heaters
Maintenance	Available
Heating Rate	20℃/min / Changeable
ETC	Customized design and fabrication suitable for customer's equipment

ACTUAL

TEMPERATURE RECORDING DATA

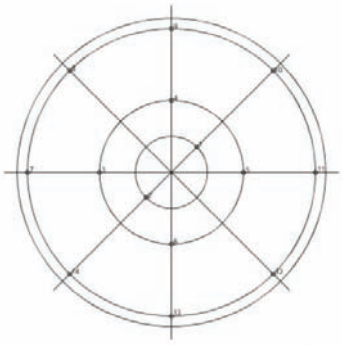




열분포 사진



- A uniform temperature
- Temperature uniformity can be adjusted within ±3℃ (based on chamber environment)



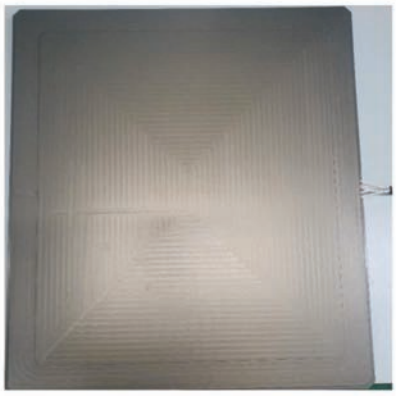
METAL HEATER

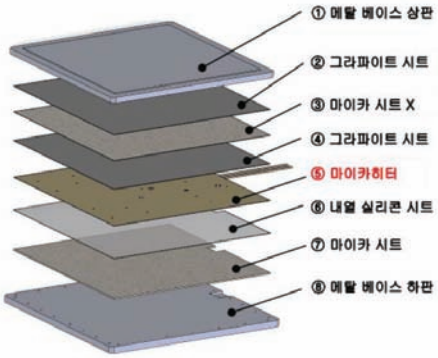


SPECIFICATIONS

Operation Temp	Up to 150℃
Production Size	750 x 650
Thickness	Not fixed
Input Voltage	AC 220V
Option	Anodizing, Suction, Sensor, Polishing etc
Quality of the material	Aluminum, Oxygen Free Copper, MMC etc Or can be used in a vacuum environment
Heater	MICA Heater
Uniformity	Within ±10℃ (at 150℃), Multi Zone control is needed (5~9zone)
Heating Rate	20℃/min / Changeable
ETC	Customized design and fabrication suitable for customer's equipment

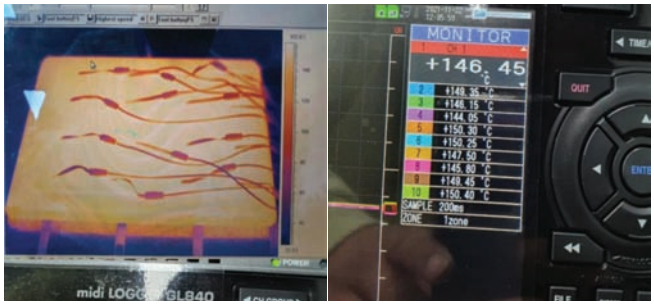
UNIFORMITY TEST







현장 test 사진



온도 분포도 사진

AIR HEATER



응용분야

항공우주
- 연소 및 흐름 시뮬레이션

섬유
- 플라스틱 또는 비닐의 접착
- 열처리가 필요한 특수 직물

포장
- 코팅 판지 실링
- 열수축 설비
- 성형
- 접착제의 강화
- 병 또는 판지의 살균

반도체/전자장비
- 웨이브 솔더 머신의 에어 나이프
- 납땜 및 납땜 제거 (PC보드, 리드 프레임, 커패시티)
- 웨어피 및 PC 보드 건조
- 와이어의 열수축을 위한 절연
- 공정에 사용되는 가스의 예열

자동차
- 열풍 접착
- 접착제의 경화
- 진공성형
- 엔진배기 시뮬레이션
- 연료전기 시험

살균
- 제약
- 의료/외과 용 하드웨어
- 포장 재료

플라스틱/고무
- 압출시 불순물 제거
- 플라스틱의 성형 및 벤딩
- 사출된 제품의 찌꺼기 제거

종이/인쇄
- 코팅된 종이의 고속건조
- 접착제의 활성화
- 잉크 건조

구조적 / 기능적 특징

- 금속의 발열선을 세라믹으로 몰딩
- 금속 발열선이 산화가 발생하지 않고, 파티클이 발생하지 않음.

성능비교

	SMG 에어 히터	타사 에어 히터
에어공급 차단 시	히터의 단선이 없음	히터의 단선 발생 가능성 有
발열선 구조	세라믹 몰딩	복잡한 열신이 공기중에 노출
	불순물에 의한 쇼트 및 화재 위험이 제거됨	불순물에 의한 쇼트 및 화재 위험이 상존함
습도	습도에 강함	습도에 취약함
작동순서	작동 순서가 단순함	작동 순서가 복잡함

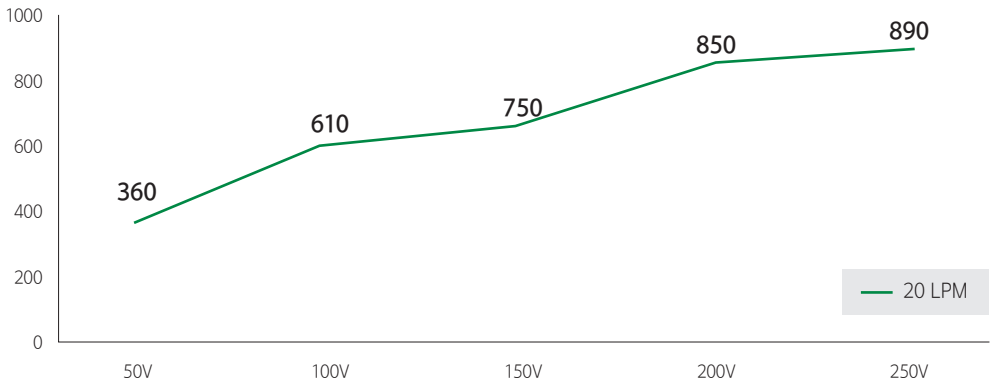
SMG 에어히터 작업 시작 및 종료

- 히터의 전원 ON/OFF의 순서가 없음
- 에어공급과 관계없이 전원 ON/OFF 가능

타사 에어히터 작업 시작 및 종료

- 작업시작 : 에어 공금 후, 히터 전원 ON
- 작업종료 : 히터 전원 OFF 후, 에어 공급 차단

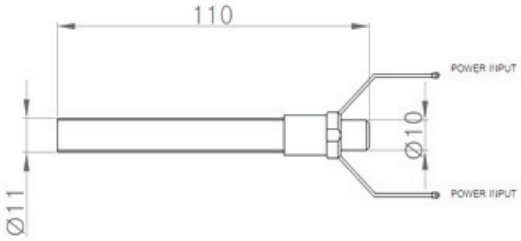
열적 성능

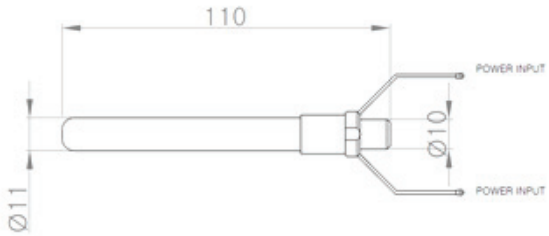


인가 전압	토출 에어 온도			
	20 LPM	30 LPM	40 LPM	50 LPM
50V	360℃	310℃	250℃	200℃
100V	610℃	580℃	520℃	400℃
150V	750℃	760℃	730℃	580℃
200V	850℃	870℃	870℃	800℃
220V	890℃	900℃	910℃	830℃









PRODUCT SPECIFICATION

Type	Circle
Indication	Air Heater
Working temperature	토출 Max 900℃
Heater body material	SUS 304 or Quartz & Ceramic
Lead wire	Depends on each product specification

CARTRIDGE HEATER 사양 및 용도 (EX)

제작 사이즈	도면 참조
Heater 사용온도	토출 Max 900℃ (센서 위치에 따라 상이함)
발열선 재질	텅스텐 or 몰리브덴
정격 전압	220 Vac
정격 출력	750W or 1000W
적용 유량	20~50 LPM
온도센서(OPTION)	열전대, Type "K"

세라믹 에어히터 / Ni-Cr 에어 히터 비교 시험

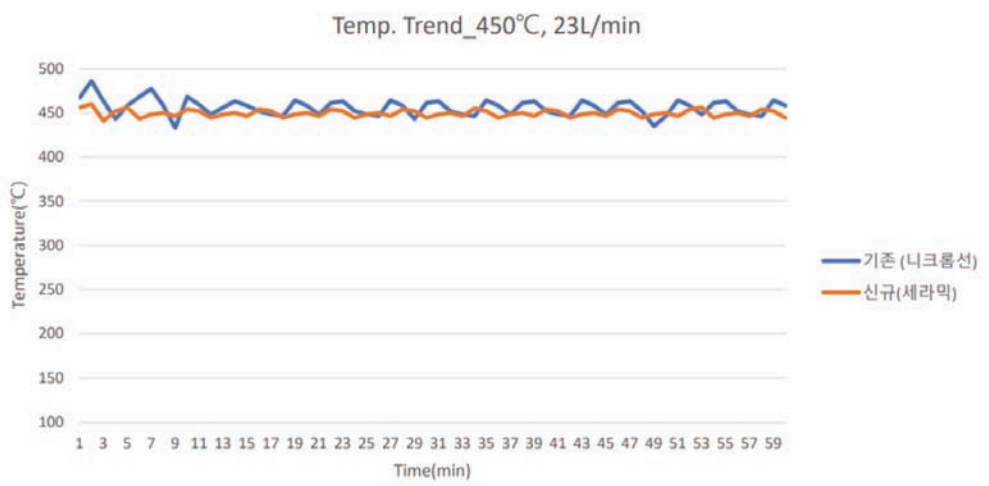
시험 목적

- 450℃, 23L/min 의 조건 하에서 대상 에어히터 간의 온도추이 확인

시험 조건

- 히터사양 : Ni-Cr히터 400W / 세라믹히터 170W
- 측정기기 : Temp. control unit
- 유지온도 : 450℃
- 실내온도 : 18℃
- 공기유량 : 23L/min
- 측정시간 : 1h

시험 결과

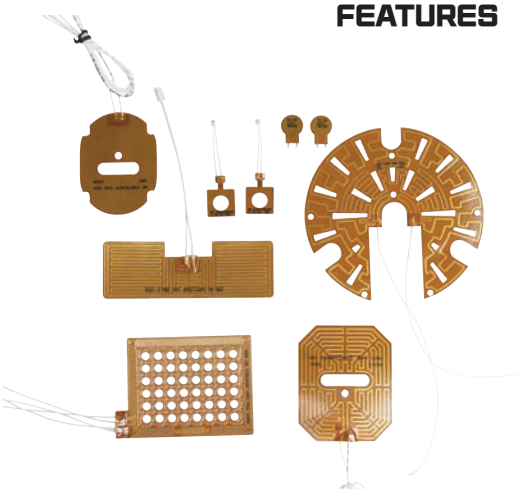


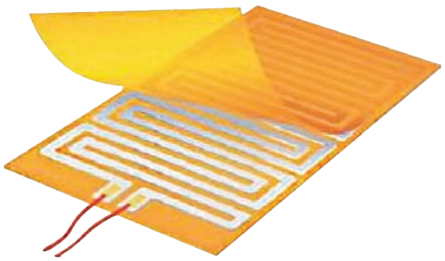
결과 검토

구분	세라믹 에어히터	Ni-Cr 에어히터
목표온도 도달시간	약 1분	약 1분 이내
온도 안정화 시간	약 5분	약 12분
안정화 후 온도 편차	Min. 443℃ / Max. 456℃	Min. 435℃ / Max. 464℃
평균 소모전류	0.74A	1.8A

HK

Kapton Film Heater





FEATURES

- 캡톤(Polymide film) 사이에 Resistance wire 또는 Etching foil 열선을 사용하여 빠른 열전달에 용이하다.
- 내화학성이 우수하고 전기적 노이즈에 강하다.
- 두께가 얇고, 높은 내열성을 지니고 있다.
- 면상발열제로 다른 Heater에 비해 균일한 온도 균일도 설계가 가능하다.
- 다양한 사이즈 및 형대로 제작이 가능하다.
- 유연성이 뛰어나 곡면에 부착이 가능하다.
- PSA(양면테이프옵션) 처리로 부착이 편리하다.
- 실리콘 접착제에 드로잉 한 저항선 방식은 150℃(상온기준)사용이 가능하다.
- FEP 접착제를 사용한 에칭 포일 방식은 200℃(상온기준)사용이 가능하다.
- 고전압 저전력 설계 실현 (소형 사이즈 제작에 용이) 가능하다.
- CCTV 설계 개발자의 설계 요구에 맞게 효율이 높다.
- 얇은 저항선(Ø0.025~ Ø0.07)을 사용하여 시야를 가리지 않고 적용 가능하다. (ex. 원형 유리부착, LCD 히팅용)
- 옥외용 CCTV 카메라의 렌즈에 김서림 방지를 목적으로 카메라 렌즈의 외부 인접 부위에 저전력 플랙시블 히터를 장착하여 김서림을 막음으로써 양질의 화상을 구현하다.
- 높은 저항 밀집도 구현이 가능하다.

- Fast heat-up and response to changing heat loads.
- Leads : Teflon insulated (MIL-W-16878), 12(300mm) long(wire gauge varies with heater)
- Other Features : The heater sheet can be ordered with or without pressure sensitive adhesive(PSA) depending on your needs.

APPLICATION

방위산업 장비, 기계, 항공기, 각종 의료 장비, 실험실 장비, 안과 장비, 사진촬영 장비, LCD 표시 장비, 컴퓨터 장비, 그 외 다수 사용처

- CCTV 유리에 부착하여 서리 제거
- CCTV 내부 회로 온도 보상 및 서리 제거
- 외부설치 CCTV 회로 구동용
- 그 외 다수 사용처

Military/aerospace, where low outgassing properties are required, Medical where through cleaning or sterilization is needed, Laboratory research, Optical equipment, Photographic equipment, LCD displays, Computer equipment.

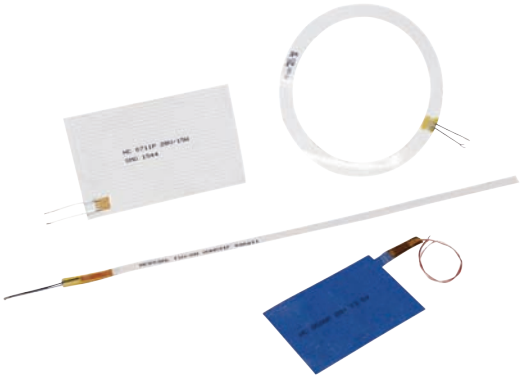
SPECIFICATION

Operating temp.	-50 ~ 160℃
Thickness	0.254mm 이내 (PSA 제외 시)
Production size	600mm × 500mm(그 외 SIZE 제작 협의)
Material	Polymide Film (적용가능 : PET, FEP, PTFE 등)
Available Input Voltage	DC/AC (상세사항 협의)
MAX temp.	200℃ (그 외 온도 및 상세사항 문의)
Maximum power density	At 25℃ in open Air 1W/㎠ (7W/in) (방열판 1~1.5T 부착 조건)

Material heating element	Resistance Wire 또는 Austenite 계 Stainless 박판 (Etching Pattern)
Insulation withstand voltage	AC 500V ~ 1000V (60 sec)
Insulation Resistance	1000Ω (500Vdc)
Resistance tolerance	상온 기준 ±10% (그 외 사양 협의)
Terminal handling	고온납 (그 외 사양 협의) 리드와이어 인출 방향 협의
Connection method	2 wire, 3 wire, 4 wire etc.

HC

Clear view Heater



FEATURES

HC 히터는 가느다란 열선을 이용한 패턴을 광학필름 사이에 넣어 빛의 차단없이 요구되는 열을 발생시킨다.

Featuring a micro-thin wire heating element laid a pattern between optical grade polyester
sheets, Clear view heaters provide reliable heat without blocking light.

APPLICATION

김서림제거용 히터, 의료용 히터, 의료장비, 자동차용 백미러, 군용노트북, 네비게이터, LED, LCD, PDA, 그 외 다수 사용처

HOW TO ORDER

Sample : HC 1020P 220V 30W

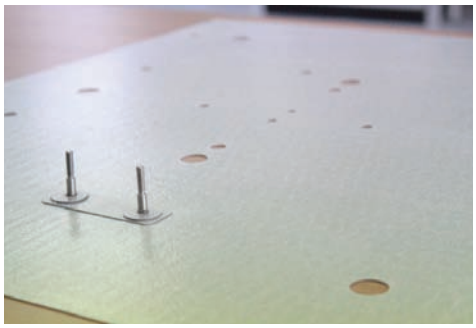
HC ..... Model
1020 ..... Size : 10mm×20mm
P ..... With PSA
220V ..... Voltage
30W ..... Wattage

SPECIFICATION

Heater material	Polyester+Metal
Operating temperature	-55℃ ~ +80℃
Transparency	82% minimum light transmission

HM, HMC

Mica Plate Heater



FEATURES

- 고온용으로 사용 가능하도록 주문 제작으로 설계된다.
- 마이카 판을 사용 환경에 따라 진공 압착 또는 접착하여 제조한다.
- 마이카 사이에 박판의 Austenite계 Stainless박판을 압착 또는 접착하여 제작한다.
- 파기열체의 표면에 견고하게 클램핑 고정 부착, 우수한 열 전달 특성 및 균일한 온도분포 특성을 가진다.
- 면상발열체로 다른 Heater에 비해 균일한 온도 균일도 설계가 가능하다.
- 내전압 및 절연내력이 우수한 형태의 전연물 접합으로 다양한 형태로 가공 및 제작된다.
- 전기적 내노이즈성이 우수하다.
- 두께가 얇고(기본 두께 1mm : 0.3mm ~ 1.5mm 제작 가능), 열 밀도가 높아 고속승, 강온이 가능하다.

APPLICATION

반도체 웨이퍼 가열용, 의료장비, 군사용 장비, LCD판넬 열처리용 장비

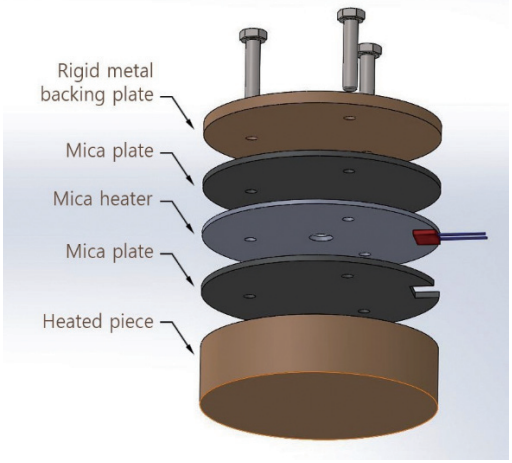
Semiconductor's wafer heater, Medical equipment, Military equipment, LCD panel Aging Equipment

HOW TO ORDER

Sample : HMC 300 A 100V 1KW

HM	Model
C	Circle
300	Size : 300 dia mm
A	With assembly
100V	Input Voltage
1KW	Wattage

mounting diagram example

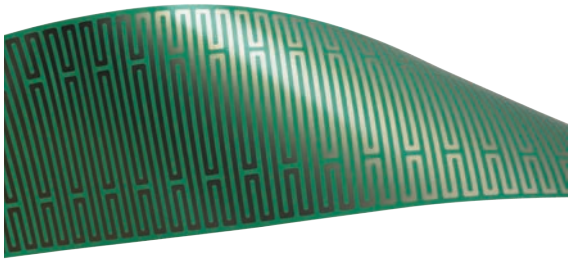


SPECIFICATION

Operating temp.	-50 ~ 500℃
Thickness	0.8mm ~ 2mm
Production size	600mm × 500mm (그 외 SIZE 제작 협의)
MAX temp.	500℃(클램핑 기준 Heater 표면 온도)
Maximum power density	At 25℃ in open Air 3W/㎠ (그 외 사양 협의)
Uniformity(100℃ 기준)	±1℃
Meterial heating element	또는 Austenite 계 Stainless 박판 (Etching Pattern)
Insulation withstand voltage	AC 1000V (60 sec)
Insulation Resistance	1000Ω (500V) (발열 온도별 상이: 협의요망)
Resistance tolerance	상온 기준 ±10% (그 외 사양 협의)
Terminal handling	브레이징(그 외 사양 협의), 리드와이어 수평 또는 수직 인출 가능
Option	Connector

HE

Epoxy Plate Heater



FEATURES

HE 히터는 에폭시 글라스판으로 절연된 히터로써 히트 싱크면에 쉽게 부착할 수 있도록 양면 접착 테이프가 부착되어 있다.

- 드로잉타입 또는 애칭타입 내장
- 유연한 에폭시 글라스판 절연구조
- 매우 얇은 형상 (0.4mm이내)
- 신속한 히팅
- 제품부착이 용이

The HE heater is insulated with an expoxy glass sheet. HE Heaters install easily on flat surfaces using tapes, double side adhesive tapes, etc.

- Drawing type or Etched foil type
- Flexible epoxy glass plate insulations
- Very thin plate
- Very fast warm up
- Pressure Sensitive Adhesive type

APPLICATION

- CCTV 카메라 가열용
- LCD 모니터의 가열용(뒷면)
- 그 외 다수 사용처

- CCTV Camera's for heating.
- LCD Monitor's heating (The back side).
- etc.

HOW TO ORDER

Sample : HE 0308P 220V 30W

HE ..... Model
0308 ..... Size : 30mm×80mm
P ..... With PSA
115V ..... Input Voltage
100W ..... Wattage

대형 HE Type Heater



SPECIFICATION

Operating temp.	-50 ~ 120℃
Thickness	0.4mm 이내 (PSA 제외)
Production size	600mm × 500mm (그 외 SIZE 제작 협의)
Material	Epoxy Glass Plate
Available Input Voltage	DC/AC (상세사항 협의)
MAX temp.	120℃ (그 외 온도 및 상세사항 문의)
Maximum power density	At 25℃ in open Air 0.65W/㎠(4W/in) (방열판 1~1.5T 부착조건)
Meterial heating element	Resistance Wire 또는 Austenite 계 Stainless 박판 (Etching Pattern)
Insulation withstand voltage	AC 500V ~ 1000V (60 sec)
Insulation Resistance	1000Ω (500Vds)
Resistance tolerance	상온 기준 ±10% (그 외 사양 협의)
Terminal handling	고온납(그 외 사양 협의), 리드와이어 인출 방향 협의
Option	PSA 양면테이프 처리, Connector Temp Sensor 내장 및 결선 (PT100, NTC, Themo couple, Thermo start)

HS

Silicone Rubber Heater



FEATURES

- 가열면 전면에 균일한 발열을 주고자 할 경우에 적합한 SILICON PAD형 HEATER
- 최고사용온도 200℃이며, 전력밀도는 0.2~0.8W/㎠
- 상하 2장의 실리콘 고무 시트의 사이에 니크롬열선을 배선하고, 내부의 공기를 제거한 후 압축 프레스 공정을 통하여 두께 1.5mm의 얇은 시트형상으로 제작됨.
- 위생 및 크린환경에 적합한 FA용 저온히터

NOTICE

- 전력밀도 0.5W/㎠ 이상인 히터는 대기중에 가열하지 말 것.(이상 발열로 인한 단선 및 발화 가능성)
- 벤딩반경 최소 R25 이상이며 히터를 휘거나 접어서 사용 불가함.
- 리드선 부분을 물, 오일 등 액체 및 습기에 접촉되지 않도록 유지할 것(누전, 단락의 원인)
- 수중, 증기, 부식성 가스 등의 상태에서는 사용불가

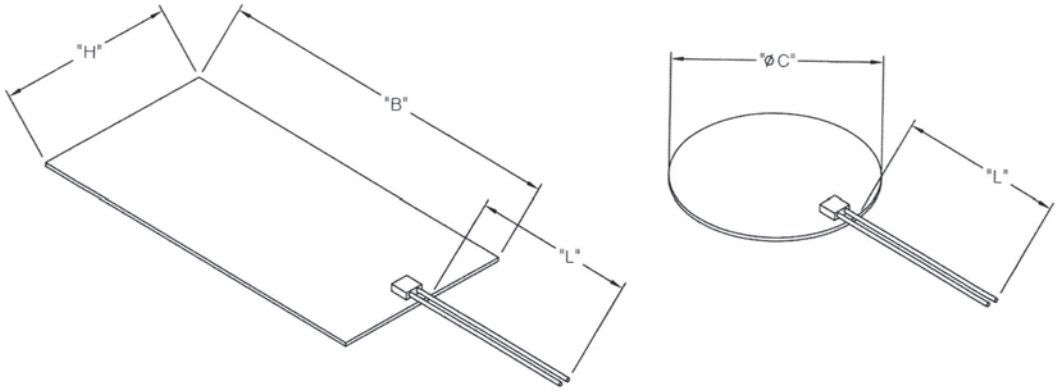
INSTALLATION

- 금속판 누름법 : 금속판으로 눌러 사용(허용 내면압력 : 15Kg/㎠)
- 실리콘 접착법 : 내열실리콘제를 도포하여 접착(최고사용온도 180℃)
- 양면테이프 접착법 : 3M 양면테잎을 붙여 접착(최고사용온도 150℃)

APPLICATION

- 동결방지 보온용
- 배관히팅 등

Order Code



ORDERING CODE

1. Model Type

HK	Kapton film type
HM	Mica plate type
HE	Epoxy plate type
HS	Silicon rubber type (SRH type)
HC	Clear PET type

2. Lead wire length

0703	70mm×30mm
1050	100mm×500mm
C300	300mm diameter
1M	1meter

3. PSA

N/A	None PSA
P	With PSA

4. Input Voltage

110	110V
220	220V

5. Wattage

30	30W

6. Lead wire length

1M	1meter

7. Option

Temp. Sensor(RTD, T/C, NTC, Thermostat), Connector

SAMPLE CODE

HK	0703	P	110	30	1M
1	2	3	4	5	6

방산부품(히터) LIST



적용처 | 레이저 경보기 성에 제거



적용처 | 조준경 및 카메라 LCD 성에 방지



적용처 | 희로발열용 히터



적용처 | 조준경 성에 방지



적용처 | 추적장치 방범용



적용처 | FM무전기 핸드셋 및 전면패널 성에 방지

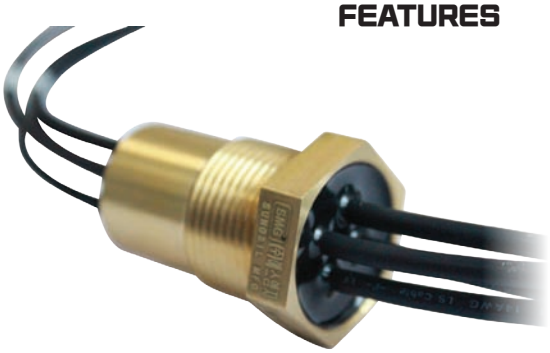


적용처 | 어뢰용 추진전지, 배터리 결빙 방지

SMG®

CB, CBC

Power, Sensor, and other lines



FEATURES

CB(Cable Bushing) 제품군은 내화성 방폭 커넥터로서, 내·외부방폭 제품의 다양한 전선(전원선, 센서선, 기타 라인 등)을 연결하는 역할을 한다.

This Cable Bushing is a flame-proof enclosure which is used for connecting various wires (power line, sensor line, and other line, etc) of internal/ external explosion-proof products.

APPLICATION

발전소, 터미널, 저장탱크, 전동기 등 방폭지역에서 사용되는 정션박스 내부의 전원 공급 및 센서 연결용

Used for power supply and sensor connections inside junction boxes in explosion-proof areas such as power plants, terminals, storage tanks, and motors.

CERTIFICATION

Complience Standard

- EN/IEC 60079-0
- EN/IEC 60079-1

Certification

- ATEX Baseefa 11ATEX 0117X/3
- IECEx BAS 11. 0056X
- KGS12-GA2BO-0245X

Operating temperature

- -55℃ ~ +110℃

Materials

- Brass

HOW TO ORDER

Sample : IECEx CB 2406 Ex d

IECEx CB
ATEX CB
KGS CB
..... Model : Cable Bushing

24 ..... Out Diameter(M24)

06 ..... Wire number of strips

Certificaion Grade ..... Ex d

SPECIFICATION

Insulation material	Epoxy
Body material	Brass or SUS
Operating temperature	-55℃ ~ +110℃
Thread size	M16 to M42

Bushing Size	M16 × 1.5	M24 × 1.5	M25 × 1.5	M32 × 1.5	M33 × 1.5	M40 × 1.5	M42 × 1.5
Lead Wire	2/4 PCS	3/4/6/8 PCS	3/4/6/8 PCS	3/6/8/12/16/24 PCS	3/6/8/12/16/24 PCS	4/6/12/24 PCS	4/6/12/24 PCS
Model Code	CB1602 CB1604A	CB2403 CB2404 CB2406A CB2408	CB2503 CB2504 CB2506A CB2508	CB3203 CB3206 CB3208 CB3212 CB3216 CB3224	CB3303 CB3306 CB3308 CB3312 CB3316 CB3324	CB4004 CB4006 CB4012 CB4024	CB4204 CB4206 CB4212 CB4224

1. Model Name

CB	Explosion proof type
CF	Nomal type

2. Type (CF model only)

S	Screw type (Standard)
H	Head + sleeve type
L	Sleeve type
F	Flange type

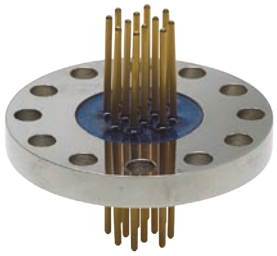
3. Size (Explosion proof type)

1602	M16×1.5, 2wire
1604A	M16×1.5, 4wire
2403	M24×1.5, 3wire
2404	M24×1.5, 4wire
2406A	M24×1.5, 6wire
2408	M24×1.5, 8wire
2503	M25×1.5, 3wire
2504	M25×1.5, 4wire
2506A	M25×1.5, 6wire
2508	M25×1.5, 8wire
3203	M32×1.5, 3wire
3206	M32×1.5, 6wire
3208	M32×1.5, 8wire
3212	M32×1.5, 12wire
3216	M32×1.5, 16wire
3224	M32×1.5, 24wire
3303	M33×1.5, 3wire
3306	M33×1.5, 6wire
3308	M33×1.5, 8wire
3312	M33×1.5, 12wire
3316	M33×1.5, 16wire
3324	M33×1.5, 24wire
4004	M40×1.5, 4wire
4006	M40×1.5, 6wire
4012	M40×1.5, 12wire
4204	M42×1.5, 4wire
4206	M42×1.5, 6wire
4212	M42×1.5, 12wire
4224	M42×1.5, 24wire

GB

Power, Sensor(RTD/TC)







FEATURES

GB(Glass Bushing/feedthrough) 제품군은 기계, 펌프, 발전기, 터빈, 제어판, 계기 및 제어 장치 등에 전원, 신호 및 데이터를 안전하게 공급하기 위한 핵심 구성 요소이다. GB Sereies는 컨덕터가 기계 및 장비가 위치한 밀폐된 공간으로 들어가는 지점에서 가스가 누출되는 것을 방지하거나, 진공을 유지하는 역할을 한다.
밀폐형 특수 유리와 금속 밀봉은 고진공도 확보 및 기밀을 유지하며, 극저온 환경에서도 사용이 가능하다. 또한, 고객의 요구 조건, 설치 환경에 따른 주문형 제품의 생산이 가능하다.

The glass bushing/feedthrough (GB) series is a key component for securely supplying power, signals, and data to machinery, pumps, generators, turbines, control panels, instruments, and controls. GB Series is responsible for preventing gases from leaking or maintaining a vacuum at the point where the conductor enters the enclosed space where machines and equipment are located.
Sealed special glass and metal sealing ensure high vacuum and maintain airtightness, and can be used even in cryogenic environments. In addition, it is possible to produce custom-made products according to customer requirements and installation environment.

APPLICATION

LNG선, LH_2선, LNG 터미널, 발전소, 반도체, 배터리 제조, 진공 시스템 등 가스 환경에서의 전원 공급 및 각종 센싱

Power supply and sensing in gaseous environments for LNG vessels, LH_2 vessels, LNG terminals, power plants, semiconductor manufacturing, battery production, vacuum systems, etc.

CERTIFICATION

Explosion proof certification

- IECEx KSCP 24.0035U, Ex eb IIC Gb

SPECIFICATIONS

Characteristics		Power type	Sensor type
Electrical rating		Up to 20 kV, 400 A	Up to 3.0 kV
Pressure resistance		Up to 100 bar	
Leakage resistance		Up to 1.0 X 10^{-11} atm·cc/sec by He leak Detector	
Temperature range		-196 ~ 100 ℃	
Materials	Pin	Cu, Sus, Ni alloy or Others	Kovar, Ni alloy or Others
	Flange	304 or 316L Stainless steel	
	Insulator	Glass	

MODEL CODES OF GB SERIES

1	Model Name		GB	Glass to metal technology
2	Outer Diameter		40	Ø40 (Ø40 ~ Ø380)
3	Number of Pin		12	12 Pin (1 ~ 120 Pin)
4	Pin Type	Power	PW020	Power Pin Ø2.0 (Ø2.0 ~ Ø20.0)
		Sensor	RT020	RTD Pin Ø2.0 (Standard)
			TK020	TC K type Ø2.0 (Ø1.4 ~ Ø2.0)
			TE020	TC E type Ø2.0 (Ø1.4 ~ Ø2.0)

SAMPLE CODE

GB		040		012		PW020
1	-	2	-	3	-	4

Note. SMG can be custom-made according to the field process.

MB

Power

FEATURES



MB(ceramic Metalizing Bushing/feedthrough) 제품군은 기계, 펌프, 발전기, 터빈 등에 전원을 안전하게 공급하기 위한 핵심 구성 요소이다. MB Sereies는 컨덕터가 기계 및 장비가 위치한 밀폐된 공간으로 들어가는 지점에서 가스가 누출되는 것을 방지하거나, 진공을 유지하는 역할을 한다.
세라믹으로 메탈라이징하면 고온에서도 안정적으로 고진공도 확보 및 기밀을 유지할 수 있다. 또한, 고객의 요구 조건, 설치 환경에 따른 주문형 제품의 생산이 가능하다.

The ceramic Metalizing Bushing/feedthrough (MB) series is a key component for securely supplying power to machinery, pumps, generators, turbines. MB Series is responsible for preventing gases from leaking or maintaining a vacuum at the point where the conductor enters the enclosed space where machines and equipment are located.
Metalizing with ceramic can stably secure high vacuum and maintain airtightness even at high temperatures. In addition, it is possible to produce custom-made products according to customer requirements and installation environment.

APPLICATION

LNG선, LH_2선, LNG 터미널, 발전소, 반도체, 배터리 제조, 진공 시스템 등 가스 환경에서의 전원 공급 및 각종 센싱

Power supply and sensing in gaseous environments for LNG vessels, LH_2 vessels, LNG terminals, power plants, semiconductor manufacturing, battery production, vacuum systems, etc.

SPECIFICATIONS

Characteristics		Power type
Electrical rating		Up to 20 kV, 400 A
Pressure resistance		Up to 40 bar
Leakage resistance		Up to 1.0 X 10^{-11} atm·cc/sec by He leak Detector
Temperature range		-196 ~ 200 ℃
Materials	Pin	Cu, Sus, Ni alloy or Others
	Flange	304 or 316L Stainless steel
	Insulator	Ceramic

MODEL CODES OF GB SERIES

1	Model Name		MB	Ceramic Metalizing technology
2	Outer Diameter		40	Ø40 (Ø40 ~ Ø380)
3	Number of Pin		6	6 Pin (1 ~ 6 Pin)
4	Pin Type	Power	PW020	Power Pin Ø2.0 (Ø2.0 ~ Ø16.0)

SAMPLE CODE

MB		040		006		PW020
1	-	2	-	3	-	4

Note. SMG can be custom-made according to the field process.

SPT

Industrial Pressure Transmitter





INTRODUCTION

SPT Pressure Transmitter is a cost-effective product which adopts microfused technology pressure sensor and specially designed amplifier circuit.
With linear correction and temperature compensation, it can effectively resist overpressure, the sensing part is made of stainless steel which can solve the problem of momentary overload pressure and also could meet different needs for pressure measurement and control in various environmental conditions. The product is batch produced in an automatic production line and all the specifications are strictly controlled. The whole transmitter experiences a strict testing for sensing element, semi-finished and completed product and aging screening, which makes it more stable and has a good consistency. It could be widely used in the fields like Io VF constant pressure water supply, compressor, T terminal, energy & water treatment system and building automation system, etc.

SPT 압력 트랜스미터는 마이크로퓨즈 기술 압력 센서와 특별히 설계된 증폭기 회로를 채택한 비용 효율적인 제품입니다. 선형 보정 및 온도 보상 기능을 통해 과압을 효과적으로 견딜 수 있으며, 감지 부분은 스테인리스 스틸로 만들어져 순간 과부하 압력 문제를 해결할 수 있으며 다양한 환경 조건에서 압력 측정 및 제어에 대한 다양한 요구 사항을 충족할 수 있습니다. 이 제품은 자동 생산 라인에서 일괄 생산되며 모든 사양은 엄격하게 제어됩니다. 전체 트랜스미터는 감지 요소, 반제품 및 완제품, 노화 스크리닝에 대한 엄격한 테스트를 거쳐 안정성이 더 높고 일관성이 좋습니다. Io VF 정압 급수, 압축기, T 단자, 에너지 및 수처리 시스템, 빌딩 자동화 시스템 등의 분야에서 널리 사용될 수 있습니다.

Features

- Micro-fused technology, dry-type structure
- Water hammer preventing and cracking resistant;
- Compact structure, high cost-effectiveness, wild range of application Temperature compensation and linear correction.

Applications

- Energy & Water Treatment System
- BuildingAutomationSystem
- Refrigeration system

STANDARD FEATURES

Power
- DC 12 ~ 28V

Pressure Range
- 0 ~ 50 / 250 / 500 and 1000 bar

Process Connection
- 1/4, 3/8 and 1/2

Accuracy
- ≤±1%F.S
  within compensation temperature
  (Optional : ±0.25%F.S)

Wetted Parts
- Stainless Steel

Overpressure
- ≤2XF.S

Output signal
- 4 ~ 20mA, 0 ~ 10V, etc.

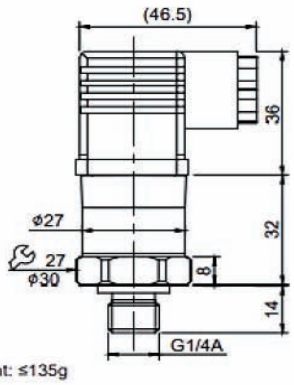
Temperature
- -40℃ ~ 100℃
  (Optional:-40℃ ~ 120℃)

Burst Pressure
- ≤5×F.S

Insulation Resistance
- IP65 (Optional IP68)

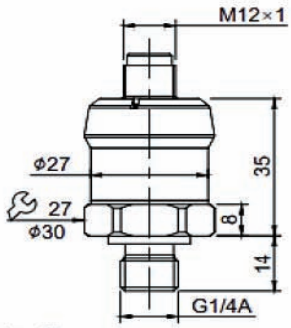
DIMENSION

Angular Connector A Type



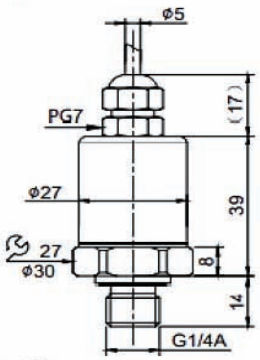
Weight: ≤135g

M12×1(4-pin) Plug Type



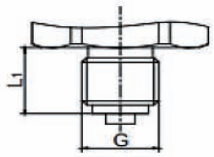
Weight: ≤95g

Cable Type

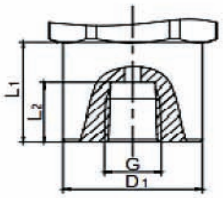


Weight: ≤160g

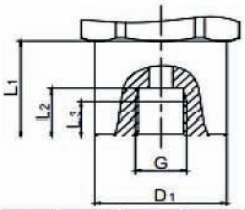
Process Connection(Unit: mm)



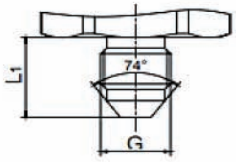
G	L1	Standard
G1/4 B	13	EN 837
G3/8 B	16	
G1/2 B	20	
M20×1.5	20	



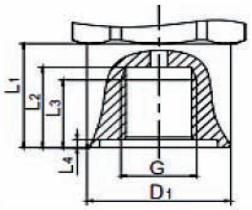
G	D1	L1	L2	Standard
NPT1/8	Φ25	20	12	ANSI/ASME B1.20.1
NPT1/4	Φ25	20	14	
NPT1/2	Φ25	25	19	



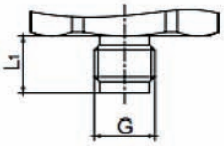
G	D1	L1	L2	L3	Standard
G1/8	Φ25	20	10	7.5	EN 837
G1/4	Φ25	20	13	10	



G	L1	Standard
7/16-20 UNF-74°Taper	14	SAE J514 E



G	D1	L1	L2	L3	Standard
G1/4	Φ25	20	15.5	13	ISO 1179-1



G	L1	Standard
G1/8 B	10	EN 837

SAMPLE CODE

SPT - XXX - X - X - X - X - X - X

Range

050 : 0 ~ 50 bar 250 : 0 ~ 250 bar 500 : 0 ~ 500 bar
1K0 : 0 ~ 1000 bar xxx : Others

Output

A : 4 ~ 20 mA K1 : 1 ~ 5 V K2 : 0 ~ 5V
K3 : 0 ~ 10V E : Others

Type

G : Gauge A : Absolute

Pressure Unit

U1 : bar U2 : Pa U3 : psi U4 : Others

Conduit Connection

M : M12 D : Din Plug
C : Cable X : Others

Process Connection

C : G1/4 C1 : G3/8
C2 : G1/2 C3 : 7/16"-20UNF-2B

Setting Range

X : Actual pressure range

TECHNICAL REFERENCE

Temperature Limit Table

Upper Temperature Limit for Various Wire Sizes in Degress ℃, (℉)						
Thermocouple Type	**8 Gage**	**14 Gage**	**20 Gage**	**24 Gage**	**28 Gage**	**30 Gage**
Units	3.25mm 0.128 in	1.63mm 0.064 in	0.81mm 0.032 in	0.51mm 0.020 in	0.33mm 0.013 in	0.25mm 0.010 in
T	N/A	370(700)	260(500)	200(400)	200(400)	150(300)
J	760(1400)	590(1100)	480(900)	370(700)	370(700)	320(600)
E	870(1600)	650(1200)	540(1000)	430(800)	430(800)	370(700)
K	1260(2300)	1090(2000)	980(1800)	870(1600)	870(1600)	760(1400)

T/C ANSI(JIS) Color Code

ANSI				
	E Chromel-Constantan Brown + Purple - Red	J Iron-Constantan Brown + White - Red	K Chromel-Alumel Brown + Yellow - Red	T Copper-Constantan Brown + Blue - Red
JIS	Purple + Red - White	Yellow + Red - White	Blue + Red - White	Brown + Red - White
Sensitivity (mV/℃)	0.063	0.053	0.041	0.043
Temperature (℃)	Milivolts			
-200	-8.824	-7.890	-5.891	-5.603
-150	-7.279	-6.499	-4.912	-4.648
-100	-5.237	-4.632	-3.553	-3.378
-50	-2.787	-2.431	-1.889	-1.819
0	0.000	0.000	0.000	0.000
50	3.047	2.585	2.022	2.035
100	6.317	5.268	4.095	4.277
150	9.787	8.008	6.137	6.702
200	13.419	10.777	8.137	9.286
250	17.178	13.553	10.151	12.011
300	21.033	16.325	12.207	14.860
350	24.961	19.089	14.292	17.816
400	28.943	21.846	16.395	20.869
450	32.960	24.607	18.513	
500	36.999	27.388	20.640	
550	41.045	30.210	22.772	

R - Platinum 13% Rhodium vs Platinum, S - Platinum 10% Rhodium vs Platinum, B - Platinum 30% Rhodium vs Platinum 6% Rhodium

International symbol for tolerance - For Resistors

Symbol	T	Q	A	B	C
Tolerance	±0.01%	±0.02%	±0.05%	±0.1%	±0.25%
Symbol	D	F	J	K	M
Tolerance	±0.5%	±1%	±5%	±10%	±20%

PT 100Ω 저항값과 온도 비교표

PT 100Ω at 0℃ 3850 PPM

-200	18.52											-200
-190	22.83	22.40	21.97	21.54	21.11	20.68	20.25	19.82	19.38	18.95	18.52	-190
-180	27.10	26.67	26.24	25.82	25.39	24.97	24.54	24.11	23.68	23.25	22.83	-180
-170	31.34	30.91	30.49	30.07	29.64	29.22	28.80	28.37	27.95	27.52	27.10	-170
-160	35.54	35.12	34.70	34.28	33.86	33.44	33.02	32.60	32.18	31.76	31.34	-160
-150	39.72	39.31	38.89	38.47	38.05	37.64	37.22	36.80	36.38	35.96	35.54	-150
-140	43.88	43.46	43.05	42.63	42.22	41.80	41.39	40.97	40.56	40.14	39.72	-140
-130	48.00	47.59	47.18	46.77	46.36	45.94	45.53	45.12	44.70	44.29	43.88	-130
-120	52.11	51.70	51.29	50.88	50.47	50.06	49.65	49.24	48.83	48.42	48.00	-120
-110	56.19	55.79	55.38	54.97	54.56	54.15	53.75	53.34	52.93	52.52	52.11	-110
-100	60.26	59.85	59.44	59.04	58.63	58.23	57.82	57.41	57.01	56.60	56.19	-100
-90	64.30	63.90	63.49	63.09	62.68	62.28	61.88	61.47	61.07	60.66	60.26	-90
-80	68.33	67.92	67.52	67.12	66.72	66.31	65.91	65.51	65.11	64.70	64.30	-80
-70	72.33	71.93	71.53	71.13	70.73	70.33	69.93	69.53	69.13	68.73	68.33	-70
-60	76.33	75.93	75.53	75.13	74.73	74.33	73.93	73.53	73.13	72.73	72.33	-60
-50	80.31	79.91	79.51	79.11	78.72	78.32	77.92	77.52	77.12	76.73	76.33	-50
-40	84.27	83.87	83.48	83.08	82.69	82.29	81.89	81.50	81.10	80.70	80.31	-40
-30	88.22	87.83	87.43	87.04	86.64	86.25	85.85	85.46	85.06	84.67	84.27	-30
-20	92.16	91.77	91.37	90.98	90.59	90.19	89.80	89.40	89.01	88.62	88.22	-20
-10	96.09	95.69	95.30	94.91	94.52	94.12	93.73	93.34	92.95	92.55	92.16	-10
0	100.00	99.61	99.22	98.83	98.44	98.04	97.65	97.26	96.87	96.48	96.09	0
0	100.00	100.39	100.78	101.17	101.56	101.95	102.34	102.73	103.12	103.51	103.90	0
10	103.90	104.29	104.68	105.07	105.46	105.85	106.24	106.63	107.02	107.40	107.79	10
20	107.79	108.18	108.57	108.96	109.35	109.73	110.12	110.51	110.90	111.29	111.67	20
30	111.67	112.06	112.45	112.83	113.22	113.61	114.00	114.38	114.77	115.15	115.54	30
40	115.54	115.93	116.31	116.70	117.08	117.47	117.86	118.24	118.63	119.01	119.40	40
50	119.40	119.78	120.17	120.55	120.94	121.32	121.71	122.09	122.47	122.86	123.24	50
60	123.24	123.63	124.01	124.39	124.78	125.16	125.54	125.93	126.31	126.69	127.08	60
70	127.08	127.46	127.84	128.22	128.61	128.99	129.37	129.75	130.13	130.52	130.90	70
80	130.90	131.28	131.66	132.04	132.42	132.80	133.18	133.57	133.95	134.33	134.71	80
90	134.71	135.09	135.47	135.85	136.23	136.61	136.99	137.37	137.75	138.13	138.51	90
100	138.51	138.88	139.26	139.64	140.02	140.40	140.78	141.16	141.54	141.91	142.29	100
110	142.29	142.67	143.05	143.43	143.80	144.18	144.56	144.94	145.31	145.69	146.07	110
120	146.07	146.44	146.82	147.20	147.57	147.95	148.33	148.70	149.08	149.46	149.83	120
130	149.83	150.21	150.58	150.96	151.33	151.71	152.08	152.46	152.83	153.21	153.58	130
140	153.58	153.96	154.33	154.71	155.08	155.46	155.83	156.20	156.58	156.95	157.33	140
150	157.33	157.70	158.07	158.45	158.82	159.19	159.56	159.94	160.31	160.68	161.05	150
160	161.05	161.43	161.80	162.17	162.54	162.91	163.29	163.66	164.03	164.40	164.77	160
170	164.77	165.14	165.51	165.89	166.26	166.63	167.00	167.37	167.74	168.11	168.48	170
180	168.48	168.85	169.22	169.59	169.96	170.33	170.70	171.07	171.43	171.80	172.17	180
190	172.17	172.54	172.91	173.28	173.65	174.02	174.38	174.75	175.12	175.49	175.86	190
200	175.86	176.22	176.59	176.96	177.33	177.69	178.06	178.43	178.79	179.16	179.53	200
210	179.53	179.89	180.26	180.63	180.99	181.36	181.72	182.09	182.46	182.82	183.19	210
220	183.19	183.55	183.92	184.28	184.65	185.01	185.38	185.74	186.11	186.47	186.84	220
230	186.84	187.20	187.56	187.93	188.29	188.66	189.02	189.38	189.75	190.11	190.47	230
240	190.47	190.84	191.20	191.56	191.92	192.29	192.65	193.01	193.37	193.74	194.10	240
250	194.10	194.46	194.82	195.18	195.55	195.91	196.27	196.63	196.99	197.35	197.71	250
260	197.71	198.07	198.43	198.79	199.15	199.51	199.87	200.23	200.59	200.95	201.31	260
270	201.31	201.67	202.03	202.39	202.75	203.11	203.47	203.83	204.19	204.55	204.90	270
280	204.90	205.26	205.62	205.98	206.34	206.70	207.05	207.41	207.77	208.13	208.48	280
290	208.48	208.84	209.20	209.56	209.91	210.27	210.63	210.98	211.34	211.70	212.05	290

PT 100Ω at 0°C 3850 PPM

300	212.05	212.41	212.76	213.12	213.48	213.83	214.19	214.54	214.90	215.25	215.61	300
310	215.61	215.96	216.32	216.67	217.03	217.38	217.74	218.09	218.44	218.80	219.15	310
320	219.15	219.51	219.86	220.21	220.57	220.92	221.27	221.63	221.98	222.33	222.68	320
330	222.68	223.04	223.39	223.74	224.09	224.45	224.80	225.15	225.50	225.85	226.21	330
340	226.21	226.56	226.91	227.26	227.61	227.96	228.31	228.66	229.02	229.37	229.72	340
350	229.72	230.07	230.42	230.77	231.12	231.47	231.82	232.17	232.52	232.87	233.21	350
360	233.21	233.56	233.91	234.26	234.61	234.96	235.31	235.66	236.00	236.35	236.70	360
370	236.70	237.05	237.40	237.74	238.09	238.44	238.79	239.13	239.48	239.83	240.18	370
380	240.18	240.52	240.87	241.22	241.56	241.91	242.26	242.60	242.95	243.29	243.64	380
390	243.64	243.99	244.33	244.68	245.02	245.37	245.71	246.06	246.40	246.75	247.09	390
400	247.09	247.44	247.78	248.13	248.47	248.81	249.16	249.50	249.85	250.19	250.53	400
410	250.53	250.88	251.22	251.56	251.91	252.25	252.59	252.93	253.28	253.62	253.96	410
420	253.96	254.30	254.65	254.99	255.33	255.67	256.01	256.35	256.70	257.04	257.38	420
430	257.38	257.72	258.06	258.40	258.74	259.08	259.42	259.76	260.10	260.44	260.78	430
440	260.78	261.12	261.46	261.80	262.14	262.48	262.82	263.16	263.50	263.84	264.18	440
450	264.18	264.52	264.86	265.20	265.53	265.87	266.21	266.55	266.89	267.22	267.56	450
460	267.56	267.90	268.24	268.57	268.91	269.25	269.59	269.92	270.26	270.60	270.93	460
470	270.93	271.27	271.61	271.94	272.28	272.61	272.95	273.29	273.62	273.96	274.29	470
480	274.29	274.63	274.96	275.30	275.63	275.97	276.30	276.64	276.97	277.31	277.64	480
490	277.64	277.98	278.31	278.64	278.98	279.31	279.64	279.98	280.31	280.64	280.98	490
500	280.98	281.31	281.64	281.98	282.31	282.64	282.97	283.31	283.64	283.97	284.30	500
510	284.30	284.63	284.97	285.30	285.63	285.96	286.29	286.62	286.95	287.29	287.62	510
520	287.62	287.95	288.28	288.61	288.94	289.27	289.60	289.93	290.26	290.59	290.92	520
530	290.92	291.25	291.58	291.91	292.24	292.56	292.89	293.22	293.55	293.88	294.21	530
540	294.21	294.54	294.86	295.19	295.52	295.85	296.18	296.50	296.83	297.16	297.49	540
550	297.49	297.81	298.14	298.47	298.80	299.12	299.45	299.78	300.10	300.43	300.75	550
560	300.75	301.08	301.41	301.73	302.06	302.38	302.71	303.03	303.36	303.69	304.01	560
570	304.01	304.34	304.66	304.98	305.31	305.63	305.96	306.28	306.61	306.93	307.25	570
580	307.25	307.58	307.90	308.23	308.55	308.87	309.20	309.52	309.84	310.16	310.49	580
590	310.49	310.81	311.13	311.45	311.78	312.10	312.42	312.74	313.06	313.39	313.71	590
600	313.71	314.03	314.35	314.67	314.99	315.31	315.64	315.96	316.28	316.60	316.92	600
610	316.92	317.24	317.56	317.88	318.20	318.52	318.84	319.16	319.48	319.80	320.12	610
620	320.12	320.43	320.75	321.07	321.39	321.71	322.03	322.35	322.67	322.98	323.30	620
630	323.30	323.62	323.94	324.26	324.57	324.89	325.21	325.53	325.84	326.16	326.48	630
640	326.48	326.79	327.11	327.43	327.74	328.06	328.38	328.69	329.01	329.32	329.64	640
650	329.64	329.96	330.27	330.59	330.90	331.22	331.53	331.85	332.16	332.48	332.79	650
660	332.79	333.11	333.42	333.74	334.05	334.36	334.68	334.99	335.31	335.62	335.93	660
670	335.93	336.25	336.56	336.87	337.18	337.50	337.81	338.12	338.44	338.75	339.06	670
680	339.06	339.37	339.69	340.00	340.31	340.62	340.93	341.24	341.56	341.87	342.18	680
690	342.18	342.49	342.80	343.11	343.42	343.73	344.04	344.35	344.66	344.97	345.28	690
700	345.28	345.59	345.90	346.21	346.52	346.83	347.14	347.45	347.76	348.07	348.38	700
710	348.38	348.69	348.99	349.30	349.61	349.92	350.23	350.54	350.84	351.15	351.46	710
720	351.46	351.77	352.08	352.38	352.69	353.00	353.30	353.61	353.92	354.22	354.53	720
730	354.53	354.84	355.14	355.45	355.76	356.06	356.37	356.67	356.98	357.28	357.59	730
740	357.59	357.90	358.20	358.51	358.81	359.12	359.42	359.72	360.03	360.33	360.64	740
750	360.64	360.94	361.25	361.55	361.85	362.16	362.46	362.76	363.07	363.37	363.67	750
760	363.67	363.98	364.28	364.58	364.89	365.19	365.49	365.79	366.10	366.40	366.70	760
770	366.70	367.00	367.30	367.60	367.91	368.21	368.51	368.81	369.11	369.41	369.71	770
780	369.71	370.01	370.31	370.61	370.91	371.21	371.51	371.81	372.11	372.41	372.71	780
790	372.71	373.01	373.31	373.61	373.91	374.21	374.51	374.81	375.11	375.41	375.70	790
800	375.70	376.00	376.30	376.60	376.90	377.19	377.49	377.79	378.09	378.39	378.68	800
810	378.68	378.98	379.28	379.57	379.87	380.17	380.46	380.76	381.06	381.35	381.65	810
820	381.65	381.95	382.24	382.54	382.83	383.13	383.42	383.72	384.01	384.31	384.60	820
830	384.60	384.90	385.19	385.49	385.78	386.08	386.37	386.67	386.96	387.25	387.55	830
840	387.55	387.84	388.14	388.43	388.72	389.02	389.31	389.60	389.90	390.19	390.48	840
850	390.48											850

PT 100Ω at 0°C 3920 PPM

-200	17.08											-200
-190	21.46	21.02	20.58	20.15	19.71	19.27	18.83	18.40	17.96	17.52	17.08	-190
-180	25.80	25.37	24.94	24.50	24.07	23.63	23.20	22.76	22.33	21.89	21.46	-180
-170	30.11	29.68	29.25	28.82	28.39	27.96	27.53	27.10	26.67	26.23	25.80	-170
-160	34.39	33.97	33.54	33.11	32.69	32.26	31.83	31.40	30.97	30.54	30.11	-160
-150	38.65	38.22	37.80	37.37	36.95	36.52	36.10	35.67	35.25	34.82	34.39	-150
-140	42.87	42.45	42.03	41.61	41.19	40.76	40.34	39.92	39.49	39.07	38.65	-140
-130	47.07	46.66	46.24	45.82	45.40	44.98	44.56	44.14	43.72	43.29	42.87	-130
-120	51.25	50.84	50.42	50.00	49.58	49.17	48.75	48.33	47.91	47.49	47.07	-120
-110	55.41	54.99	54.58	54.16	53.75	53.33	52.92	52.50	52.09	51.67	51.25	-110
-100	59.54	59.13	58.72	58.30	57.89	57.48	57.06	56.65	56.24	55.82	55.41	-100
-90	63.66	63.25	62.84	62.43	62.01	61.60	61.19	60.78	60.37	59.96	59.54	-90
-80	67.76	67.35	66.94	66.53	66.12	65.71	65.30	64.89	64.48	64.07	63.66	-80
-70	71.84	71.43	71.02	70.61	70.21	69.80	69.39	68.98	68.57	68.17	67.76	-70
-60	75.90	75.50	75.09	74.68	74.28	73.87	73.47	73.06	72.65	72.24	71.84	-60
-50	79.95	79.55	79.14	78.74	78.33	77.93	77.52	77.12	76.71	76.31	75.90	-50
-40	83.99	83.58	83.18	82.78	82.38	81.97	81.57	81.16	80.76	80.36	79.95	-40
-30	88.01	87.61	87.21	86.80	86.40	86.00	85.60	85.20	84.79	84.39	83.99	-30
-20	92.02	91.62	91.22	90.82	90.42	90.02	89.61	89.21	88.81	88.41	88.01	-20
-10	96.02	95.62	95.22	94.82	94.42	94.02	93.62	93.22	92.82	92.42	92.02	-10
0	100.00	99.60	99.20	98.81	98.41	98.01	97.61	97.21	96.81	96.41	96.02	0
0	100.00	100.40	100.80	101.19	101.59	101.99	102.39	102.78	103.18	103.58	103.97	0
10	103.97	104.37	104.77	105.16	105.56	105.95	106.35	106.75	107.14	107.54	107.93	10
20	107.93	108.33	108.72	109.12	109.52	109.91	110.30	110.70	111.09	111.49	111.88	20
30	111.88	112.28	112.67	113.07	113.46	113.85	114.25	114.64	115.03	115.43	115.82	30
40	115.82	116.21	116.61	117.00	117.39	117.79	118.18	118.57	118.96	119.35	119.75	40
50	119.75	120.14	120.53	120.92	121.31	121.71	122.10	122.49	122.88	123.27	123.66	50
60	123.66	124.05	124.44	124.83	125.22	125.61	126.00	126.39	126.78	127.17	127.56	60
70	127.56	127.95	128.34	128.73	129.12	129.51	129.90	130.29	130.68	131.07	131.45	70
80	131.45	131.84	132.23	132.62	133.01	133.39	133.78	134.17	134.56	134.95	135.33	80
90	135.33	135.72	136.11	136.49	136.88	137.27	137.65	138.04	138.43	138.81	139.20	90
100	139.20	139.59	139.97	140.36	140.74	141.13	141.51	141.90	142.29	142.67	143.06	100
110	143.06	143.44	143.83	144.21	144.59	144.98	145.36	145.75	146.13	146.52	146.90	110
120	146.90	147.28	147.67	148.05	148.43	148.82	149.20	149.58	149.97	150.35	150.73	120
130	150.73	151.11	151.50	151.88	152.26	152.64	153.02	153.41	153.79	154.17	154.55	130
140	154.55	154.93	155.31	155.70	156.08	156.46	156.84	157.22	157.60	157.98	158.36	140
150	158.36	158.74	159.12	159.50	159.88	160.26	160.64	161.02	161.40	161.78	162.16	150
160	162.16	162.54	162.91	163.29	163.67	164.05	164.43	164.81	165.19	165.56	165.94	160
170	165.94	166.32	166.70	167.07	167.45	167.83	168.21	168.58	168.96	169.34	169.71	170
180	169.71	170.09	170.47	170.84	171.22	171.60	171.97	172.35	172.73	173.10	173.48	180
190	173.48	173.85	174.23	174.60	174.98	175.35	175.73	176.10	176.48	176.85	177.23	190
200	177.23	177.60	177.97	178.35	178.72	179.10	179.47	179.84	180.22	180.59	180.96	200
210	180.96	181.34	181.71	182.08	182.46	182.83	183.20	183.57	183.95	184.32	184.69	210
220	184.69	185.06	185.43	185.81	186.18	186.55	186.92	187.29	187.66	188.03	188.41	220
230	188.41	188.78	189.15	189.52	189.89	190.26	190.63	191.00	191.37	191.74	192.11	230
240	192.11	192.48	192.85	193.22	193.59	193.96	194.32	194.69	195.06	195.43	195.80	240
250	195.80	196.17	196.54	196.90	197.27	197.64	198.01	198.38	198.74	199.11	199.48	250
260	199.48	199.85	200.21	200.58	200.95	201.31	201.68	202.05	202.41	202.78	203.15	260
270	203.15	203.51	203.88	204.24	204.61	204.98	205.34	205.71	206.07	206.44	206.80	270
280	206.80	207.17	207.53	207.90	208.26	208.63	208.99	209.35	209.72	210.08	210.45	280
290	210.45	210.81	211.17	211.54	211.90	212.26	212.63	212.99	213.35	213.72	214.08	290

PT 100Ω at 0°C 3920 PPM

300	214.08	214.44	214.80	215.17	215.53	215.89	216.25	216.61	216.98	217.34	217.70	300
310	217.70	218.06	218.42	218.78	219.14	219.51	219.87	220.23	220.59	220.95	221.31	310
320	221.31	221.67	222.03	222.39	222.75	223.11	223.47	223.83	224.19	224.55	224.91	320
330	224.91	225.26	225.62	225.98	226.34	226.70	227.06	227.42	227.78	228.13	228.49	330
340	228.49	228.85	229.21	229.56	229.92	230.28	230.64	230.99	231.35	231.71	232.07	340
350	232.07	232.42	232.78	233.13	233.49	233.85	234.20	234.56	234.92	235.27	235.63	350
360	235.63	235.98	236.34	236.69	237.05	237.40	237.76	238.11	238.47	238.82	239.18	360
370	239.18	239.53	239.89	240.24	240.59	240.95	241.30	241.66	242.01	242.36	242.72	370
380	242.72	243.07	243.42	243.78	244.13	244.48	244.83	245.19	245.54	245.89	246.24	380
390	246.24	246.59	246.95	247.30	247.65	248.00	248.35	248.70	249.06	249.41	249.76	390
400	249.76	250.11	250.46	250.81	251.16	251.51	251.86	252.21	252.56	252.91	253.26	400
410	253.26	253.61	253.96	254.31	254.66	255.01	255.36	255.71	256.06	256.40	256.75	410
420	256.75	257.10	257.45	257.80	258.15	258.49	258.84	259.19	259.54	259.89	260.23	420
430	260.23	260.58	260.93	261.27	261.62	261.97	262.31	262.66	263.01	263.35	263.70	430
440	263.70	264.05	264.39	264.74	265.08	265.43	265.78	266.12	266.47	266.81	267.16	440
450	267.16	267.50	267.85	268.19	268.54	268.88	269.23	269.57	269.91	270.26	270.60	450
460	270.60	270.95	271.29	271.63	271.98	272.32	272.66	273.01	273.35	273.69	274.03	460
470	274.03	274.38	274.72	275.06	275.40	275.75	276.09	276.43	276.77	277.11	277.46	470
480	277.46	277.80	278.14	278.48	278.82	279.16	279.50	279.84	280.18	280.52	280.87	480
490	280.87	281.21	281.55	281.89	282.23	282.57	282.91	283.24	283.58	283.92	284.26	490
500	284.26	284.60	284.94	285.28	285.62	285.96	286.30	286.63	286.97	287.31	287.65	500
510	287.65	287.99	288.32	288.66	289.00	289.34	289.67	290.01	290.35	290.69	291.02	510
520	291.02	291.36	291.70	292.03	292.37	292.71	293.04	293.38	293.71	294.05	294.39	520
530	294.39	294.72	295.06	295.39	295.73	296.06	296.40	296.73	297.07	297.40	297.74	530
540	297.74	298.07	298.41	298.74	299.07	299.41	299.74	300.07	300.41	300.74	301.08	540
550	301.08	301.41	301.74	302.07	302.41	302.74	303.07	303.41	303.74	304.07	304.40	550
560	304.40	304.73	305.07	305.40	305.73	306.06	306.39	306.72	307.06	307.39	307.72	560
570	307.72	308.05	308.38	308.71	309.04	309.37	309.70	310.03	310.36	310.69	311.02	570
580	311.02	311.35	311.68	312.01	312.34	312.67	313.00	313.33	313.66	313.99	314.31	580
590	314.31	314.64	314.97	315.30	315.63	315.96	316.28	316.61	316.94	317.27	317.59	590
600	317.59	317.92	318.25	318.58	318.90	319.23	319.56	319.88	320.21	320.54	320.86	600
610	320.86	321.19	321.52	321.84	322.17	322.49	322.82	323.14	323.47	323.79	324.12	610
620	324.12	324.44	324.77	325.09	325.42	325.74	326.07	326.39	326.72	327.04	327.36	620
630	327.36	327.69	328.01	328.34	328.66	328.98	329.31	329.63	329.95	330.28	330.60	630
640	330.60	330.92	331.24	331.57	331.89	332.21	332.53	332.85	333.18	333.50	333.82	640
650	333.82	334.14	334.46	334.78	335.11	335.43	335.75	336.07	336.39	336.71	337.03	650
660	337.03											660

PT 1000Ω/NI 120Ω 저항값과 온도 비교표

PT 1000Ω at 0℃ 3850 PPM

-50	803.10	799.10	795.10	791.10	787.20	783.20	779.20	775.20	771.30	767.30	763.30
-40	842.70	838.80	834.80	830.80	826.90	822.90	818.90	815.00	811.00	807.00	803.10
-30	882.20	878.30	874.30	870.40	866.40	862.50	858.50	854.60	850.60	846.70	842.70
-20	921.60	917.70	913.70	909.80	905.90	901.90	898.00	894.00	890.10	886.20	882.20
-10	960.90	956.90	953.00	949.10	945.20	941.20	937.30	933.40	929.50	925.50	921.60
-0	1000.00	996.10	992.20	988.30	984.40	980.40	976.50	972.60	968.70	964.80	960.90
℃	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
0	1000.00	1003.90	1007.80	1011.70	1015.60	1019.50	1023.40	1027.30	1031.20	1035.10	1039.00
10	1039.00	1042.90	1046.80	1050.70	1054.60	1058.50	1062.40	1066.30	1070.20	1074.00	1077.90
20	1077.90	1081.80	1085.70	1089.60	1093.50	1097.30	1101.20	1105.10	1109.00	1112.80	1116.70
30	1116.70	1120.60	1124.50	1128.30	1132.20	1136.10	1139.90	1143.80	1147.70	1151.50	1155.40
40	1155.40	1159.30	1163.10	1167.00	1170.80	1174.70	1178.50	1182.40	1186.20	1190.10	1194.00
50	1194.00	1197.80	1201.60	1205.50	1209.30	1213.20	1217.00	1220.90	1224.70	1228.60	1232.40
60	1232.40	1236.20	1240.10	1243.90	1247.70	1251.60	1255.40	1259.20	1263.10	1266.90	1270.70
70	1270.70	1274.50	1278.40	1282.20	1286.00	1289.80	1293.70	1297.50	1301.30	1305.10	1308.90
80	1308.90	1312.70	1316.60	1320.40	1324.20	1328.00	1331.80	1335.60	1339.40	1343.20	1347.00
90	1347.00	1350.80	1354.60	1358.40	1362.20	1366.00	1369.80	1373.60	1377.40	1381.20	1385.00
100	1385.00	1388.80	1392.60	1396.40	1400.20	1403.90	1407.70	1411.50	1415.30	1419.10	1422.90
110	1422.90	1426.60	1430.40	1434.20	1438.00	1441.70	1445.50	1449.30	1453.10	1456.80	1460.60
120	1460.60	1464.40	1468.10	1471.90	1475.70	1479.40	1483.20	1487.00	1490.70	1494.50	1498.20
130	1498.20	1502.00	1505.70	1509.50	1513.30	1517.00	1520.80	1524.50	1528.30	1532.00	1535.80
140	1535.80	1539.50	1543.20	1547.00	1550.70	1554.50	1558.20	1561.90	1565.70	1569.40	1573.10
150	1573.10	1576.90	1580.60	1584.30	1588.10	1591.80	1595.50	1599.30	1603.00	1606.70	1610.40

NI 120Ω at 0℃ 6720 PPM

-70	73.1	72.45	71.8	71.15	70.5	69.85	69.2	68.55	67.9	67.25
-60	79.62	78.97	78.31	77.66	77.01	76.36	75.71	75.06	74.4	73.75
-50	86.17	85.51	84.86	84.2	83.55	82.89	82.24	81.65	80.93	80.27
-40	92.76	92.1	91.44	90.78	90.12	89.46	88.8	88.14	87.48	86.83
-30	99.41	98.74	98.07	97.41	96.74	96.07	95.41	94.75	94.08	93.42
-20	106.15	105.47	104.79	104.11	103.44	102.77	102.09	101.42	100.75	100.08
-10	113	112.31	111.62	110.93	110.25	109.56	108.87	108.19	107.51	106.83
-0	120	119.29	118.65	117.88	117.17	116.47	115.47	115.08	114.38	113.69
℃	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
0	120	120.71	121.42	122.13	122.85	123.57	124.38	125	125.72	126.45
10	127.17	127.9	128.63	129.36	130.09	130.82	131.56	132.29	133.03	133.77
20	134.65	135.26	136.01	136.76	137.51	138.26	139.02	139.77	140.53	141.29
30	142.06	142.82	143.44	144.36	145.13	145.9	145.68	147.46	148.23	149.01
40	149.79	150.65	151.36	165.15	165.94	153.74	154.53	155.33	156.13	156.93
50	157.74	165.55	144.36	160.17	160.98	161.8	162.62	163.44	164.26	165.08
60	165.9	166.73	167.56	168.38	169.21	170.05	170.88	171.72	172.56	173.4
70	174.25	175.1	175.95	176.8	177.66	178.51	179.37	180.24	181.1	181.97
80	182.84	183.71	184.44	185.46	186.34	187.22	188.1	188.98	189.87	190.75
90	191.64	192.53	193.42	194.32	195.21	196.11	197.01	197.91	198.82	199.73
100	200.64	201.55	202.46	203.38	204.3	205.22	206.14	207.06	207.99	208.92
110	209.85	210.78	211.72	212.66	213.6	214.54	215.49	216.44	217.39	218.34
120	219.29	220.25	221.21	222.17	223.14	224.1	225.07	226.04	227.01	227.99
130	228.96	229.94	230.92	231.9	232.89	233.88	234.87	235.86	236.85	237.85
140	238.85	239.85	240.85	241.86	242.86	243.87	244.88	245.89	246.91	247.93
150	248.95	249.97	251	265.02	253.05	254.09	255.12	256.16	257.21	265.25
160	244.3	260.35	261.4	262.45	263.51	264.57	265.63	266.6	267.76	268.83
170	269.91	270.98	272.06	273.14	274.22	275.3	276.39	277.48	278.57	279.67
180	280.77	281.87	282.98	284.09	285.21	286.33	287.45	288.57	289.7	290.83
190	291.96	293.1	294.24	295.38	296.65	297.67	298.82	299.97	301.13	302.29
200	303.46	304.62	305.8	306.97	308.15	309.34	310.65	311.72	312.91	314.11
210	315.31	316.65	317.72	318.94	320.15	321.37	322.44	323.82	325.05	326.29
220	327.53	328.77	330.02	331.27	332.65	333.78	335.05	336.32	337.44	338.86
230	340.14	341.42	342.71	344	345.3	346.44	347.9	349.2	350.51	351.82
240	353.14	354.46	355.79	357.12	365.45	344.79	361.13	362.47	363.82	365.18
250	366.53	367.89	369.26	370.62	372	373.37	374.75	376.13	377.65	378.91
260	380.31	381.7	383.11	384.65	385.93	387.34	388.77	390.19	391.62	393.05
270	394.49	395.93	379.38	398.82	400.28	401.73	403.19	404.66	406.12	407.6
280	409.07	410.55	412.03	413.65	415.01	416.51	418.01	419.51	421.02	422.53
290	424.05	425.57	427.09	428.62	430.16	431.7	433.24	434.78	436.33	437.89
300	439.44									

CU 10Ω 저항값과 온도 비교표

CU 10Ω at 25℃ 4270 PPM

-90	5.526	5.486	5.446	5.407	5.367	5.327	5.287	5.247	5.208	5.168
-80	5.923	5.883	5.844	5.804	5.764	5.725	5.685	5.645	5.606	5.566
-70	6.318	6.279	6.239	6.2	6.16	6.121	6.081	6.042	6.002	5.962
-60	6.712	6.672	6.633	6.594	6.554	6.515	6.476	6.436	6.397	6.37
-50	7.104	7.064	7.025	6.986	6.947	6.908	6.869	6.83	6.79	6.751
-40	7.49	7.451	7.413	7.374	7.335	7.296	7.27	7.22	7.181	7.142
-30	7.876	7.838	7.799	7.761	7.722	7.683	7.645	7.606	7.568	7.529
-20	8.263	8.224	8.185	8.147	8.108	8.07	8.031	7.992	7.954	7.915
-10	8.649	8.61	8.572	8.533	8.494	8.456	8.417	8.378	8.34	8.301
-0	9.035	8.996	8.97	8.919	8.881	8.842	8.805	8.77	8.726	8.687
℃	0	1	2	3	4	5	6	7	8	9
0	9.035	9.074	9.112	9.151	9.189	9.228	9.267	9.305	9.344	9.383
10	9.421	9.46	9.498	9.537	9.576	9.614	9.703	9.692	9.73	9.769
20	9.807	9.846	9.885	9.923	9.962	10	10.039	10.078	10.116	10.155
30	10.194	10.232	10.271	10.309	10.348	10.387	10.425	10.464	10.502	10.541
40	10.7	10.618	10.707	10.696	10.734	10.773	10.811	10.85	10.889	10.927
50	10.966	11.005	11.043	11.082	11.12	11.159	11.198	11.236	11.275	11.313
60	11.352	11.391	11.429	11.468	11.507	11.545	11.704	11.622	11.661	11.7
70	11.738	11.777	11.816	11.854	11.863	11.931	11.97	12.009	12.074	12.086
80	12.124	12.163	12.202	12.24	12.279	12.318	12.356	12.395	12.433	12.472
90	12.511	12.549	12.708	12.67	12.67	12.704	12.742	12.781	12.82	12.87
100	12.897	12.935	12.974	13.013	13.051	13.09	13.129	13.167	13.206	13.244
110	13.283	13.322	13.36	13.399	13.437	13.476	13.515	13.553	13.592	13.631
120	13.669	13.708	13.746	13.785	13.824	13.862	13.901	13.94	13.978	14.017
130	14.055	14.094	14.133	14.171	14.21	14.248	14.287	14.326	14.364	14.403
140	14.442	14.48	14.519	14.557	14.596	14.635	14.673	14.712	14.751	14.789
150	14.828	14.867	14.906	14.945	14.984	15.022	15.061	15.1	15.139	15.178
160	15.217	15.256	15.295	15.334	15.373	15.412	15.451	15.49	15.529	15.568
170	15.607	15.646	15.685	15.724	15.763	15.802	15.84	15.879	15.918	15.957
180	15.996	16.035	16.074	16.113	16.17	16.191	16.23	16.269	16.308	16.347
190	16.386	16.425	16.464	16.503	16.542	16.701	16.62	16.709	16.698	16.737
200	16.776	16.815	16.854	16.893	16.932	16.971	17.01	17.049	17.088	17.127
210	17.166	17.205	17.244	17.283	17.321	17.36	17.399	17.438	17.477	17.516
220	17.555	17.594	17.633	17.672	17.711	17.75	19.789	17.828	17.867	17.906
230	17.945	17.984	18.023	18.062	18.101	18.14	18.179	18.218	18.257	18.296
240	18.335	18.374	18.413	18.452	18.491	18.53	18.569	18.608	18.648	18.687
250	18.726	18.77	18.804	18.843	18.882	18.921	18.96	18.999	19.038	19.077
260	19.116									

CU 10Ω at 0℃ 4270 PPM

℃	0	1	2	3	4	5	6	7	8	9
0	10.000	10.043	10.336	10.129	10.172	10.215	10.257	10.388	18.343	18.386
10	10.429	10.472	10.515	10.558	10.601	10.644	10.687	10.730	18.773	18.815
20	10.350	10.901	10.944	10.937	11.860	11.073	11.116	11.159	11.202	11.245
30	11.288	11.331	11.374	11.417	11.460	11.503	11.546	11.589	11.632	11.675
40	11.718	11.761	11.384	11.347	11.890	11.933	11.976	12.019	12.862	12.185
50	12.148	12.191	12.234	12.277	12.320	12.363	12.486	12.449	12.492	12.535
60	12.573	12.621	12.664	12.787	12.758	12.793	12.336	12.879	12.922	12.965
70	13.888	13.051	13.094	13.137	13.188	13.223	13.266	13.309	13.352	13.395
80	13.438	13.481	13.525	13.568	13.611	13.654	13.697	13.743	13.783	13.826
90	13.869	13.912	13.955	13.998	14.041	14.885	14.123	14.171	14.214	14.257
100	14.388	14.343	14.336	14.429	14.472	14.516	14.559	14.682	14.645	14.633
110	14.731	14.774	14.817	14.868	14.984	14.947	14.998	15.033	15.876	15.119
120	15.162	15.206	15.249	15.292	15.335	15.378	15.421	15.464	15.588	15.551
130	15.594	15.637	15.680	15.723	15.767	15.818	15.853	15.896	15.439	15.932
140	16.826	16.869	16.112	16.155	16.198	16.242	16.285	16.328	16.371	16.414
150	16.458	16.581	16.544	16.587	16.638	16.674	16.717	16.768	16.803	16.846
160	16.398	16.933	16.976	17.019	17.362	17.186	17.149	17.192	17.235	17.279
170	17.322	17.365	17.488	17.452	17.495	17.538	17.531	17.625	17.668	17.711
180	17.754	17.798	17.841	17.884	17.927	17.971	18.814	18.857	18.181	18.144
190	18.137	18.238	18.274	18.317	18.368	18.404	18.447	18.498	18.333	18.577
200	18.628	18.663	18.787	18.758	18.793	18.837	18.880	18.923	18.966	19.818

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주식회사 에스엠지
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대표전화. 031-504-7766 **팩스.** 031-504-0200
이메일. sungsil@sungsil.com

SMG Co., Ltd.
305, Sihwa venture-ro, Siheung-si, Gyeonggi-do, South Korea
Head Office. 82-31-504-7766 **FAX.** 82-31-504-0200
E-mail. sungsil@sungsil.com

www.**sungsil**.com

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